Contexts and pseudocontexts

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In the projection lattice of a finite-dimensional Hilbert space, an orthogonal basis—a *context*—is characterised by the property that its atomic probabilities sum to one on every state. We study a weakening of this condition in which two tuples of atoms induce the same probability sum on every state. Such pairs are called *pseudocontexts*. In finite dimension this is equivalent to two distinct rank-one decompositions of the same positive operator, placing the notion naturally among finite frame-theoretic constructions. Size $k = 1$ is trivial; size $k = 2$ already admits genuinely nonorthogonal examples over $\mathbb{C}$; and, in fact, examples exist for every $k \geq 2$. For $k \geq 3$, genuine pseudocontexts also exist over $\mathbb{R}$. We present the operator formulation, give a general explicit construction, analyse the spectral structure of the associated structure operator, illustrate the symmetric $k = 3$ case with a figure, and list the remaining classification problems.

Keywords: projection lattice, pseudocontext, rank-one decomposition, structure operator, quantum contextuality

I. INTRODUCTION

Let L be the lattice of projections in a Hilbert space of finite dimension $n \geq 2$. Its atoms (minimal non-zero elements) correspond to one-dimensional subspaces, represented by nonzero vectors. States are described by probability measures $P \colon L \to [0,1]$. We denote by $S(L)$ the set of all states on L.

Vectors $a_1, \dots, a_n$ form a *context* if they compose an orthogonal basis. In this case (and only in this case) they satisfy

$$\forall P \in S(L): \quad \sum_{i=1}^{n} P(a_i) = 1. \tag{1}$$

The present note concerns weaker additive identifications obtained by comparing two tuples of atoms that are not required to be orthogonal. Such *pseudocontexts* are quantum-logical shadows of orthomodular-lattice constructions of Mayet–Navara–Rogalewicz [1] and have direct physical meaning: they encode state-independent empirical equivalences between families of generally noncommuting observables. It is then natural to ask for which sizes k and over which scalar fields genuine pseudocontexts exist. The answer is simple: pseudocontexts already exist for $k = 2$ over $\mathbb{C}$, and, more generally, for every $k \geq 2$; for $k \geq 3$ they exist over $\mathbb{R}$ as well. Table I summarises the existence landscape used below; "genuine" is understood in the supporting-subspace sense made precise in Definition 3.

II. PRELIMINARIES

Throughout, H is a Hilbert space of finite dimension $n \geq 2$, and $L = L(H)$ is the orthomodular lattice of its closed subspaces. Its atoms correspond to one-dimensional subspaces, or equivalently to *rays* spanned by nonzero vectors. We identify an atom with any representative unit vector a and write $P_a = |a\rangle\langle a|$ for the orthogonal projection onto the ray it spans. More generally, for nonzero a we set $P_a = |a\rangle\langle a| / \langle a, a\rangle$, so that P_a depends only on the ray $\mathbb{C}a$.

A *state* is a map $P \colon L \to [0,1]$ with $P(I) = 1$ that is finitely additive on orthogonal families. In dimension $n \geq 3$, Gleason's theorem [2] asserts that every state arises from a density operator ρ via

$$P(a) = \operatorname{tr}(\rho P_a), \qquad \rho \geq 0, \quad \operatorname{tr}\rho = 1. \tag{2}$$

For $n = 2$, Gleason's theorem fails for the abstract definition of a state, and we adopt (2) as the definition: a state is the assignment $P_a \mapsto \operatorname{tr}(\rho P_a)$ for some density operator ρ. Under either reading, the key consequence we use repeatedly is the following elementary separation fact.

Lemma 1 (Separation by density operators)**.** *Let A and B be self-adjoint operators on a finite-dimensional real or complex Hilbert space. If*

$$\operatorname{tr}(\rho A) = \operatorname{tr}(\rho B)$$

for every density operator ρ, then $A = B$.

Proof. Apply the hypothesis to $C := A - B$. For every unit vector x, the rank-one projection P_x is a density operator, hence

$$0 = \operatorname{tr}(P_x C) = \langle x, Cx\rangle.$$

By homogeneity the quadratic form $x \mapsto \langle x, Cx\rangle$ vanishes for every vector x. The usual polarization identity (real or complex, according to the scalar field) then implies $\langle x, Cy\rangle = 0$ for all x, y. Therefore $C = 0$, and so $A = B$. ☐

III. CONTEXTS AND PSEUDOCONTEXTS

The operator form of (1) is elementary and is the starting point for everything that follows.

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TABLE I. Existence landscape for genuine pseudocontexts.

Setting	Size	Status	Mechanism or obstruction
$\mathbb{R}$ or $\mathbb{C}$, $n \geq 2$	$k = 1$	none	One rank-one probability determines the ray (Proposition 7).
$\mathbb{C}^n$, $n \geq 2$	$k = 2$	exists	Complex phase freedom gives inequivalent decompositions (Proposition 9).
$\mathbb{R}^n$, $n \geq 2$	$k = 2$	none genuine	Real two-term decompositions are unique, or are contexts on their support (Proposition 14).
$\mathbb{R}^n$, $n \geq 2$	$k \geq 3$	exists	Planar doubled-angle zero-centroid configurations embed in higher dimension (Theorem 12).
Full support in $\mathbb{R}^n$	$k > n$	tight examples exist	Rotate a unit-norm tight frame with $T = (k/n)I_n$ (Proposition 53).

Proposition 2 (Contexts)**.** *Unit vectors $a_1, \dots, a_n \in H$ form an orthonormal basis if and only if*

$$\sum_{i=1}^{n} P_{a_i} = I. \tag{3}$$

Proof. The forward implication is immediate. For the converse, fix j and evaluate (3) on the vector a_j:

$$1 = \langle a_j, I a_j \rangle = \sum_{i=1}^{n} |\langle a_i, a_j \rangle|^2.$$

The term with $i = j$ already contributes 1, so all others must vanish. Hence $\langle a_i, a_j \rangle = 0$ for $i \neq j$. As this holds for every j, the family is orthonormal, and being of size n in an n-dimensional space it is an orthonormal basis. □

Combined with (2), Proposition 2 recovers (1) as a state-theoretic characterisation of orthogonal bases.

Definition 3 (Pseudocontext)**.** Let $k \geq 1$. A *pseudocontext of size k* is a pair of k-tuples of atoms $(a_1, \dots, a_k)$, $(b_1, \dots, b_k)$ such that

(i) each tuple contains pairwise distinct atoms (i.e. no two of the a_i are scalar multiples, and likewise for the b_i),

(ii) the two tuples are disjoint as sets of rays, so that $|\{a_1, \dots, a_k, b_1, \dots, b_k\}| = 2k$, and

(iii) $\forall P \in S(L): \quad \sum_{i=1}^{k} P(a_i) = \sum_{i=1}^{k} P(b_i).$

A pseudocontext is called *genuine* if neither k-tuple is a context on its supporting subspace, i.e. if the rays in neither tuple form an orthonormal basis of their span.

Remark 4. Condition (i) excludes repeated atoms within a single tuple, and condition (ii) ensures the two tuples are truly distinct as collections of rays. Without condition (ii), every pseudocontext of size k would yield pseudocontexts of every size $k' > k$ by adjoining $k' - k$ shared atoms; this trivial inflation can be carried out within the original ambient subspace. Condition (ii) rules out such padding and makes the size k a genuine invariant of the construction.

Remark 5 (Contexts on supporting subspaces). Accordingly, in the examples below the word “context” is often used in the relative sense of an orthonormal basis of the supporting subspace on which the construction lives. The ambient Hilbert space may be larger.

By (2) and Lemma 1, condition (iii) of Definition 3 is equivalent to the operator identity

$$T := \sum_{i=1}^{k} P_{a_i} = \sum_{i=1}^{k} P_{b_i}, \tag{4}$$

which we call the *structure operator* of the pseudocontext. Thus pseudocontexts are precisely pairs of distinct unit-norm rank-one decompositions of a single positive operator. This places them naturally among finite frame-theoretic constructions: contexts are the special case in which the structure operator is the identity on the relevant subspace; in general, pseudocontexts are unit-norm rank-one frame decompositions of a common positive operator. If the common operator is proportional to the identity on its support, the family is a tight frame; after normalisation on that support one obtains a POVM.

Remark 6 (Connection to SIC-POVMs). If the common structure operator satisfies $T = \frac{k}{n} I$ and $k = n^2$, then a single family $\{P_{a_1}, \dots, P_{a_k}\}$ scaled by n/k is a SIC-POVM exactly when the frame vectors are equiangular. Pseudocontexts are more general: they concern two distinct unit-norm decompositions of the same positive operator, without requiring equiangularity or the SIC cardinality $k = n^2$.

IV. THE SMALLEST CASES

We begin by proving the first three entries of Table I: size $k = 1$ is impossible, size $k = 2$ exists over $\mathbb{C}$, and size $k = 2$ has no genuine real realisation.

A. No pseudocontexts of size $k = 1$

Proposition 7 (Size $k = 1$)**.** *There is no pseudocontext of size $k = 1$.*

Proof. The condition $P(a_1) = P(b_1)$ for all $P \in S(L)$ means that the two self-adjoint operators P_{a_1} and P_{b_1} have equal expectation in every density operator. By Lemma 1, $P_{a_1} = P_{b_1}$. This rank-one identity forces a_1 and b_1 to span the same ray, contradicting the distinctness condition (ii) of Definition 3. □

B. Pseudocontexts of size $k = 2$ and the Bloch representation

In dimension two the pseudocontext condition admits a complete characterisation in terms of Bloch vectors. This yields a unified and basis-independent description of all pseudocontexts and clarifies the real–complex dichotomy.

Every rank-one projector on $\mathbb{C}^2$ can be written as

$$P_a = \tfrac{1}{2}\big(I + \vec{r}_a \cdot \vec{\sigma}\big), \qquad \vec{r}_a \in \mathbb{R}^3, \quad \|\vec{r}_a\| = 1, \tag{5}$$

where $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices. To be specific, for any unit vector $a = \alpha e_1 + \beta e_2 \in \mathbb{C}^2$ in a chosen orthonormal basis (e_1, e_2), the components of the associated Bloch vector $\vec{r}_a = (r_x, r_y, r_z)$ are obtained by computing the expectation values of the Pauli matrices:

$$\begin{aligned} r_x &= \langle a, \sigma_x a\rangle = 2\,\mathrm{Re}(\alpha^* \beta), \\ r_y &= \langle a, \sigma_y a\rangle = 2\,\mathrm{Im}(\alpha^* \beta), \\ r_z &= \langle a, \sigma_z a\rangle = |\alpha|^2 - |\beta|^2. \end{aligned} \tag{6}$$

Proposition 8 (Bloch representation of pseudocontexts in dimension $d = 2$)**.** *Let $(a_1, \ldots, a_k)$ and $(b_1, \ldots, b_k)$ be two tuples of unit vectors in $\mathbb{C}^2$ satisfying the distinctness/disjointness conditions of Definition 3(i)–(ii). Then the following are equivalent:*

1. *condition (iii) of Definition 3 holds;*
2. $\sum_{i=1}^{k} P_{a_i} = \sum_{i=1}^{k} P_{b_i}$;
3. $\sum_{i=1}^{k} \vec{r}_{a_i} = \sum_{i=1}^{k} \vec{r}_{b_i}$.

The common structure operator T has Bloch form

$$T = \tfrac{k}{2} I + \tfrac{1}{2} \vec{R} \cdot \vec{\sigma}, \qquad \vec{R} = \sum_{i=1}^{k} \vec{r}_{a_i} = \sum_{i=1}^{k} \vec{r}_{b_i}.$$

Proof. By Definition 3 and (4), conditions 1 and 2 are equivalent. For the equivalence with 3, substitute the Bloch representation (5) into the structure operator:

$$\sum_{i=1}^{k} P_{a_i} = \sum_{i=1}^{k} \tfrac{1}{2}\big(I + \vec{r}_{a_i} \cdot \vec{\sigma}\big) = \tfrac{k}{2} I + \tfrac{1}{2}\Big(\sum_{i=1}^{k} \vec{r}_{a_i}\Big) \cdot \vec{\sigma}.$$

Since $I, \sigma_x, \sigma_y, \sigma_z$ are linearly independent, equality of the two structure operators is equivalent to equality of the Bloch-vector sums. $\square$

C. Explicit $k = 2$ example via Bloch vectors

We can now construct an explicit $k = 2$ pseudocontext by directly applying the Bloch representation and showing exactly how it arises from non-orthogonal configurations.

Proposition 9 (Existence of $k = 2$ complex pseudocontexts)**.** *There are genuinely nonorthogonal pseudocontexts of size $k = 2$ in $\mathbb{C}^n$ for every $n \geq 2$.*

Proof. Embed $\mathbb{C}^2$ into $\mathbb{C}^n$ as the span of the first two basis vectors e_1, e_2. Define two pairs of unit vectors:

$$u_\pm = \Big(\tfrac{\sqrt{3}}{2}, \pm\tfrac{1}{2}, 0, \ldots, 0\Big), \qquad v_\pm = \Big(\tfrac{\sqrt{3}}{2}, \pm\tfrac{i}{2}, 0, \ldots, 0\Big).$$

We obtain their Bloch vectors explicitly using (6). For $u_\pm$, setting $\alpha = \sqrt{3}/2$ and $\beta = \pm 1/2$, we compute:

$$\begin{aligned} r_x &= 2\,\mathrm{Re}\big(\tfrac{\sqrt{3}}{2} \cdot (\pm\tfrac{1}{2})\big) = \pm\tfrac{\sqrt{3}}{2}, \\ r_y &= 2\,\mathrm{Im}\big(\tfrac{\sqrt{3}}{2} \cdot (\pm\tfrac{1}{2})\big) = 0, \\ r_z &= \big|\tfrac{\sqrt{3}}{2}\big|^2 - \big|\pm\tfrac{1}{2}\big|^2 = \tfrac{3}{4} - \tfrac{1}{4} = \tfrac{1}{2}. \end{aligned}$$

Thus $\vec{r}_{u_\pm} = (\pm\frac{\sqrt{3}}{2}, 0, \frac{1}{2})$, which lie in the xz-plane of the Bloch sphere.

For $v_\pm$, setting $\alpha = \sqrt{3}/2$ and $\beta = \pm i/2$, we have $\alpha^* \beta = (\sqrt{3}/2)(\pm i/2)$, so:

$$\begin{aligned} r_x &= 2\,\mathrm{Re}(\tfrac{\sqrt{3}}{2} \cdot \pm\tfrac{i}{2}) = 0, \\ r_y &= 2\,\mathrm{Im}(\tfrac{\sqrt{3}}{2} \cdot \pm\tfrac{i}{2}) = \pm\tfrac{\sqrt{3}}{2}, \\ r_z &= \tfrac{3}{4} - \tfrac{1}{4} = \tfrac{1}{2}. \end{aligned}$$

Thus $\vec{r}_{v_\pm} = (0, \pm\frac{\sqrt{3}}{2}, \frac{1}{2})$, which lie in the yz-plane.

Summing the Bloch vectors within each pair yields the same structure vector:

$$\vec{R} = \vec{r}_{u_+} + \vec{r}_{u_-} = \vec{r}_{v_+} + \vec{r}_{v_-} = (0, 0, 1).$$

By Proposition 8, this guarantees that the operator sums match in the two-dimensional subspace. Embedding back into $\mathbb{C}^n$ gives, with $V := \mathrm{span}\{e_1, e_2\}$,

$$\begin{aligned} P_{u_+} + P_{u_-} &= P_{v_+} + P_{v_-} = \tfrac{3}{2} P_{e_1} + \tfrac{1}{2} P_{e_2} \\ &= \big(I_V + \tfrac{1}{2}\sigma_z\big) \oplus 0_{n-2} = \begin{pmatrix} 3/2 & 0 \\ 0 & 1/2 \end{pmatrix} \oplus 0_{n-2}. \end{aligned}$$

The four rays are distinct: u_+ and u_- differ by a sign in the real second coordinate; v_+ and v_- differ by a sign in the imaginary second coordinate. No $u_\pm$ is collinear with any $v_\pm$: comparison of the first coordinate forces the proportionality scalar to be 1, but then the second coordinates are different. Within each pair, $\langle u_+, u_-\rangle = \frac{1}{2} \neq 0$ and $\langle v_+, v_-\rangle = \frac{1}{2} \neq 0$, so neither pair is orthogonal. The pseudocontext is therefore genuine. $\square$

Remark 10 (Bloch geometry of the complex $k = 2$ example). This construction directly illustrates the continuous $U(1)$ freedom present in complex spaces. The pair of Bloch vectors $\{\vec{r}_{v_+}, \vec{r}_{v_-}\}$ is related to $\{\vec{r}_{u_+}, \vec{r}_{u_-}\}$ by a 90° rotation about the z-axis (which aligns with their common sum $\vec{R}$). Because this rotation preserves $\vec{R}$, it produces a distinct, valid decomposition of the same fixed structure operator T.

D. Classification in $\mathbb{C}^2$

Pseudocontexts in dimension two correspond exactly to distinct decompositions of a fixed vector $\vec{R} \in \mathbb{R}^3$ into k unit vectors.

Theorem 11 (Complete characterisation in $\mathbb{C}^2$)**.** *Let $k \geq 1$.*

(i) For $k = 1$, no pseudocontext exists.

(ii) For $k = 2$, pseudocontexts exist and form continuous families.

(iii) For all $k \geq 3$, pseudocontexts exist; if $\vec{R} = 0$, they arise from zero-centroid configurations (e.g. regular polygons).

Proof. Part (i) is Proposition 7.

For (ii), fix $\vec{R} \in \mathbb{R}^3$ with $0 < \|\vec{R}\| < 2$. Writing

$$\vec{r}_1 = \vec{R}/2 + \vec{w}, \qquad \vec{r}_2 = \vec{R}/2 - \vec{w},$$

the condition $\|\vec{r}_1\| = \|\vec{r}_2\| = 1$ is equivalent to $\vec{w} \perp \vec{R}$ and $\|\vec{w}\|^2 = 1 - \|\vec{R}\|^2/4$. Thus $\vec{w}$ ranges over a circle in the plane orthogonal to $\vec{R}$, so the set of unordered pairs $\{\vec{r}_1, \vec{r}_2\}$ is a continuous family.

For (iii), choose any $k \geq 3$ and take a regular k-gon on a great circle of the Bloch sphere. Let $(a_1, \dots, a_k)$ be its corresponding rays in $\mathbb{C}^2$. Let $(b_1, \dots, b_k)$ be obtained by rotating the polygon in the same great circle through an angle δ which is not a multiple of $2\pi/k$. Then no rotated vertex coincides with an original vertex, so the two ray sets are disjoint. Both k-tuples have centroid 0, hence the same structure operator, and because $k \geq 3$ neither tuple is a context. Therefore they form a genuine pseudocontext. □

E. Real–complex dichotomy in two dimensions

The $k = 2$ complex example in Proposition 9 is the simplest continuous family. In $\mathbb{R}^2$, no genuine pseudocontext of size $k = 2$ exists, as stated in Theorem 12(ii) below.

Theorem 12 (Real vs. complex in $d = 2$)**.**

(i) In $\mathbb{C}^2$, genuine pseudocontexts of size $k = 2$ exist.

(ii) In $\mathbb{R}^2$, no genuine pseudocontext of size $k = 2$ exists.

(iii) In $\mathbb{R}^2$, genuine pseudocontexts exist for all $k \geq 3$.

Proof. Part (i) follows from Proposition 9.

For (ii), represent a real ray by a unit vector $x_\theta = (\cos\theta, \sin\theta)$, with angles understood modulo π. Its projector is

$$P_\theta = \frac{1}{2}\begin{pmatrix} 1 + \cos 2\theta & \sin 2\theta \\ \sin 2\theta & 1 - \cos 2\theta \end{pmatrix}.$$

Thus a real ray is parametrised by the doubled-angle unit vector $q_\theta = (\cos 2\theta, \sin 2\theta)$, and equality $P_{\theta_1} + P_{\theta_2} = P_{\phi_1} + P_{\phi_2}$ is equivalent to $q_{\theta_1} + q_{\theta_2} = q_{\phi_1} + q_{\phi_2}$. If this common sum is nonzero, a two-term decomposition of it into unit vectors in the plane is unique up to permutation, so the two unordered pairs of rays coincide. If the common sum is zero, then $q_{\theta_2} = -q_{\theta_1}$, hence the two corresponding rays are orthogonal; the pair is a context on its two-dimensional support. In either case no genuine real pseudocontext of size 2 exists.

For (iii), choose distinct real rays with angles

$$\theta_j = \theta_0 + \frac{\pi j}{k}, \qquad j = 0, \dots, k-1.$$

Their doubled-angle vectors form a regular k-gon and therefore have centroid 0. A second copy rotated by an angle $\delta \not\equiv m\pi/k \pmod{\pi}$ for all $m \in \mathbb{Z}$ is disjoint from the first one and has the same centroid. The two projector sums are consequently equal, and since $k \geq 3$ neither tuple is an orthonormal basis of its two-dimensional support. Hence the two tuples form a genuine pseudocontext. □

Remark 13 (Earlier observation). The nonexistence of genuine real pseudocontexts of size $k = 2$ in dimension two was observed in [3] without noticing that the situation is different in complex spaces. The doubled-angle (real Bloch-circle) parametrisation makes the real–complex dichotomy transparent: over $\mathbb{R}$ a decomposition of a fixed structure operator into two rank-one terms is unique up to permutation, whereas over $\mathbb{C}$ a continuous phase degree of freedom produces inequivalent decompositions.

Proposition 14. *No genuine pseudocontext of size $k = 2$ exists over a real Hilbert space of any finite dimension.*

Proof. Suppose

$$P_{a_1} + P_{a_2} = P_{b_1} + P_{b_2} =: T$$

with all vectors real and unit. The range of T is

$$\operatorname{span}\{a_1, a_2\} = \operatorname{span}\{b_1, b_2\},$$

so all four rays lie in a subspace of dimension at most two. If this subspace is one-dimensional, distinctness fails. Hence we may restrict to a real two-dimensional support.

If $a_1 \perp a_2$, then T acts as the identity on its 2-dimensional support. Proposition 2, applied within that support, then implies that any two-vector unit decomposition of the identity is an orthonormal basis of the support; so both pairs are contexts and the pseudocontext is not genuine.

If a_1 and a_2 are not orthogonal, choose their signs so that $\gamma = \langle a_1, a_2 \rangle \in (0,1)$. Then T has the simple eigenvalues $1+\gamma$ and $1-\gamma$, with eigenvectors proportional to $a_1 + a_2$ and $a_1 - a_2$. These spectral data determine the unordered pair of rays $\{[a_1], [a_2]\}$ uniquely. Hence any other real decomposition of T into two unit rank-one projections gives the same two rays up to permutation, contradicting the disjointness requirement. Therefore no genuine real size-two pseudocontext exists. □

F. Spectral form

Proposition 15 (Spectrum in $d = 2$)**.** *The eigenvalues of T are*

$$\lambda_\pm = \tfrac{k}{2} \pm \tfrac{1}{2}\|\vec{R}\|.$$

Remark 16. The vector $\vec{R}$ completely parametrises pseudocontexts in dimension two.

V. PSEUDOCONTEXTS OF EVERY SIZE $k \geq 2$

The $k = 2$ case is best understood via the Bloch-sphere parametrisation; the general k-construction below is its root-of-unity extension. This section supplies the complex existence row of Table I for all $k \geq 2$. The construction below uses a continuous phase $e^{i\phi}$ and is therefore intrinsically complex; the corresponding existence statement over $\mathbb{R}$ for $k \geq 3$ is covered separately by Example 20 in Section VI.

Theorem 17 (Complex-phase construction)**.** *Let H contain an orthonormal pair e_1, e_2. Fix an integer $k \geq 2$, let $\omega = e^{2\pi i/k}$, choose $\alpha \in (0, \pi/2)$, and choose $\phi \in \mathbb{R}$ such that $e^{i\phi}$ is not a kth root of unity. Define unit vectors*

$$a_j = \cos\alpha\, e_1 + \sin\alpha\, \omega^j e_2,$$
$$b_j = \cos\alpha\, e_1 + \sin\alpha\, e^{i\phi} \omega^j e_2, \qquad j = 0, \ldots, k-1.$$

Then the $2k$ rays spanned by the a_j and b_j are pairwise distinct, and

$$\sum_{j=0}^{k-1} P_{a_j} = \sum_{j=0}^{k-1} P_{b_j} = k\big(\cos^2\alpha\, P_{e_1} + \sin^2\alpha\, P_{e_2}\big).$$

Hence $(a_0, \ldots, a_{k-1})$ and $(b_0, \ldots, b_{k-1})$ form a pseudocontext of size k. The pseudocontext is genuine whenever $k \geq 3$ or $\alpha \neq \pi/4$.

Proof. Write $c = \cos\alpha$ and $s = \sin\alpha$. For each j,

$$P_{a_j} = c^2 P_{e_1} + s^2 P_{e_2} + cs\big(\omega^{-j}|e_1\rangle\langle e_2| + \omega^j|e_2\rangle\langle e_1|\big),$$
$$P_{b_j} = c^2 P_{e_1} + s^2 P_{e_2} + cs\big(e^{-i\phi}\omega^{-j}|e_1\rangle\langle e_2| + e^{i\phi}\omega^j|e_2\rangle\langle e_1|\big).$$

Summing over $j = 0, \ldots, k-1$ and using $\sum_{j=0}^{k-1} \omega^j = 0$ cancels all off-diagonal terms, giving

$$\sum_{j=0}^{k-1} P_{a_j} = \sum_{j=0}^{k-1} P_{b_j} = k\big(c^2 P_{e_1} + s^2 P_{e_2}\big).$$

Distinctness. If $a_j = \lambda a_\ell$ for some scalar λ, then comparing first coordinates (which equal $c \neq 0$) gives $\lambda = 1$, and then $\omega^j = \omega^\ell$, so $j = \ell$. Hence the a_j are pairwise distinct rays; the identical argument applies to the b_j. If $a_j = \lambda b_\ell$, then $\lambda = 1$ and $\omega^j = e^{i\phi}\omega^\ell$, i.e. $e^{i\phi} = \omega^{j-\ell}$, contradicting the assumption that $e^{i\phi}$ is not a kth root of unity. Hence no a_j coincides with any b_ℓ.

Genuineness. For $j \neq \ell$ within the a-tuple,

$$\langle a_j, a_\ell\rangle = c^2 + s^2\omega^{\ell-j}.$$

This vanishes only if $c^2 = s^2$ (i.e. $\alpha = \pi/4$) and $\omega^{\ell-j} = -1$. Thus, if $\alpha \neq \pi/4$, no two distinct vectors in the a-tuple are orthogonal. An identical computation gives $\langle b_j, b_\ell\rangle = c^2 + s^2\omega^{\ell-j}$, so the same conclusion holds for the b-tuple. Hence neither tuple is a context when $\alpha \neq \pi/4$. If instead $k \geq 3$, each tuple consists of k distinct rays inside the two-dimensional support $\operatorname{span}\{e_1, e_2\}$, and therefore cannot be an orthonormal basis of that support. Hence the pseudocontext is genuine whenever $k \geq 3$ or $\alpha \neq \pi/4$. □

Remark 18 (Complex versus real constructions). Theorem 17 uses a genuinely complex phase. The real examples in Example 20 and Section XI are different mechanisms and should not be conflated with the root-of-unity family.

Remark 19. At the special value $\alpha = \pi/4$, the structure operator becomes $\frac{k}{2} I_{\operatorname{span}\{e_1,e_2\}}$, so the construction yields a unit-norm tight frame on the two-dimensional support. For $k = 2$ this reduces to a context; for $k \geq 3$ it gives a highly symmetric pseudocontext (cf. Remark 21). In dimension $d = 2$, this construction reduces to the Bloch-vector description of Sec. IV B.

VI. A SYMMETRIC $k = 3$ EXAMPLE

The case $k = 3$ admits a very transparent planar representation; it is the finite-dimensional analogue of the Mercedes–Benz frame.

Example 20 (Mercedes–Benz type pseudocontext)**.** Fix a two-dimensional subspace $V \subset H$ with orthonormal basis (e_1, e_2), and fix an angle θ satisfying $\theta \not\equiv 0$, $\theta \not\equiv \pi/3$, and $\theta \not\equiv 2\pi/3$ modulo π. Put, for $j = 0, 1, 2$,

$$a_j = \cos\Big(\tfrac{2\pi j}{3}\Big)e_1 + \sin\Big(\tfrac{2\pi j}{3}\Big)e_2,$$

$$b_j = \cos\Big(\tfrac{2\pi j}{3} + \theta\Big)e_1 + \sin\Big(\tfrac{2\pi j}{3} + \theta\Big)e_2.$$

Then the six rays are pairwise distinct, and

$$\sum_{j=0}^{2} P_{a_j} = \sum_{j=0}^{2} P_{b_j} = \tfrac{3}{2} I_V.$$

Thus (a_0, a_1, a_2) and (b_0, b_1, b_2) form a genuine pseudocontext of size $k = 3$ over $\mathbb{R}$.

Proof. In the basis (e_1, e_2), each projector has the form

$$P_{\cos\varphi\, e_1 + \sin\varphi\, e_2} = \begin{pmatrix} \cos^2\varphi & \cos\varphi\sin\varphi \\ \cos\varphi\sin\varphi & \sin^2\varphi \end{pmatrix}.$$

For $\varphi_j = 2\pi j/3$ and for $\varphi_j = \theta + 2\pi j/3$, the trigonometric identities

$$\sum_{j=0}^{2} \cos(2\varphi_j) = 0, \qquad \sum_{j=0}^{2} \sin(2\varphi_j) = 0,$$

which hold for any arithmetic progression of three angles spaced $2\pi/3$ apart, imply that the off-diagonal terms cancel while the diagonal terms each sum to $3/2$. Hence both sums equal $\frac{3}{2} I_V$.

The three excluded residue classes modulo π account for all collisions between the two triples:

- $\theta \equiv 0 \pmod{\pi}$: every b_j coincides with a_j or $-a_j$, giving the same ray.
- $\theta \equiv \pi/3 \pmod{\pi}$: the b-triple is a cyclic permutation of the a-triple as a set of rays.

- $\theta \equiv 2\pi/3 \pmod{\pi}$: the b-triple is again a permutation of the a-triple.

For all other values of θ, the six rays are pairwise distinct. Genuineness holds because for generic θ no two vectors within the same triple are orthogonal (the angle between consecutive vectors in each triple is $2\pi/3 \neq \pi/2$), so neither triple is a context. $\square$

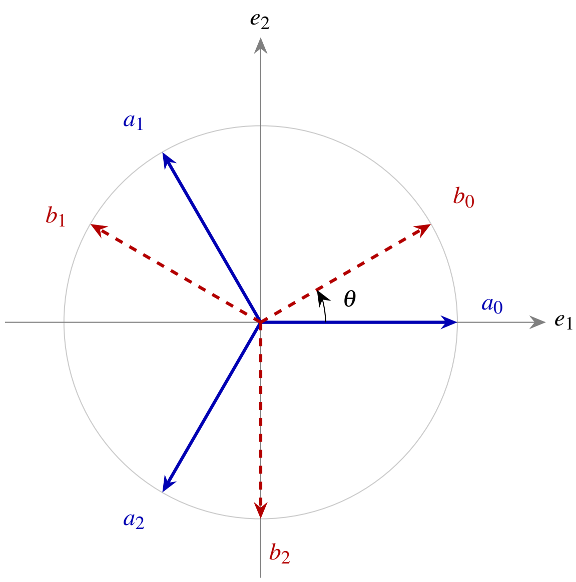

FIG. 1. The Mercedes–Benz pseudocontext of Example 20, drawn for $\theta = \pi/6$. The solid blue rays a_0, a_1, a_2 and the dashed red rays b_0, b_1, b_2 are two distinct unordered triples in the plane V, yet the associated rank-one projectors have the same sum $\frac{3}{2}I_V$.

Remark 21 (Complex trine description). Example 20 has a compact complex analogue. Let $\omega = e^{2\pi i/3}$ and embed V into $\mathbb{C}^n$ as $\mathrm{span}\{e_1, e_2\}$. The *trines*

$$u_j = \tfrac{1}{\sqrt{2}}\big(1, \omega^j, 0, \ldots, 0\big),\ v_j = \tfrac{1}{\sqrt{2}}\big(1, e^{i\phi}\omega^j, 0, \ldots, 0\big),$$

with $j = 0, 1, 2$, yield another rank-one decomposition of the same structure operator $\frac{3}{2}I_V$ for any $\phi \in \mathbb{R}$ that is not a multiple of $2\pi/3$. The identity $1 + \omega + \omega^2 = 0$ makes the off-diagonal entries of $\sum_j P_{u_j}$ and $\sum_j P_{v_j}$ cancel, giving the common structure operator $\frac{3}{2}I_V$. This form is most natural in frame-theoretic discussions and in connection with the Weyl–Heisenberg group.

Remark 22 (Minimal ambient dimension). The explicit constructions in Sections IV B, V, and VI all live in the two-dimensional subspace $\mathrm{span}\{e_1, e_2\}$, so they embed into any $\mathbb{C}^n$ or $\mathbb{R}^n$ with $n \geq 2$. This is a statement about the constructions exhibited here, not about all pseudocontexts in all dimensions; indeed, the rotational construction of Section XI produces real pseudocontexts whose rays span all of $\mathbb{R}^n$. No pseudocontext exists in dimension $n = 1$, since there is only one ray.

Remark 23 (Nondegenerate structure operators). The trine family is highly symmetric: its structure operator is a scalar multiple of the identity on its support. More complicated examples with nondegenerate structure operators occur already in dimension three. The main difference is that the pseudocontext in Example 20 has only one constant value in the probability sum, while the advanced constructions admit a larger range of values, still respecting the equality. Explicit three-dimensional implementations of nondegenerate structure operators were derived in [4].

- **15-atom hypergraph (T_1).** A pseudocontext of size $k = 3$ yields $\lambda_1 = 2$, $\lambda_2 = \frac{7+\sqrt{21}}{14} \approx 0.827$, $\lambda_3 = \frac{7-\sqrt{21}}{14} \approx 0.173$, with $\mathrm{tr}(T_1) = 3$ and $\lambda_{\max} = 2 > 1$, consistent with Proposition 28.
- **36-atom hypergraph (T_2).** A larger true-implies-false gadget gives $T_2 = \mathrm{diag}(1/5, 1/5, 13/5)$ with $\mathrm{tr}(T_2) = 3$ and $\lambda_{\max} = 13/5 > 1$.

Such examples are useful when one wants to move beyond purely symmetric frame orbits.

VII. SPECTRAL STRUCTURE AND CANONICAL FORMS

In dimension $d = 2$, all spectral properties reduce to the Bloch form of Sec. IV B. We now analyse the general properties of the structure operator $T = \sum_{i=1}^{k} P_{a_i}$ associated with a k-tuple of unit vectors $(a_1, \ldots, a_k)$ in H.

A. Gram representation and spectral equivalence

Definition 24 (Analysis operator and Gram matrix)**.** Let $A\colon \mathbb{C}^k \to H$ be defined by $Ae_i = a_i$, where (e_i) is the standard basis of $\mathbb{C}^k$. The *analysis operator* is $A^\dagger\colon H \to \mathbb{C}^k$, and the *Gram matrix* is

$$G := A^\dagger A = \big(\langle a_i, a_j\rangle\big)_{i,j=1}^{k}.$$

Proposition 25 (Frame operator factorisation)**.** *The structure operator admits the factorisation $T = AA^\dagger$, and $G = A^\dagger A$.*

Proposition 26 (Spectral equivalence)**.** *The nonzero spectra of T and G coincide (including multiplicities):*

$$\sigma(T) \setminus \{0\} = \sigma(G) \setminus \{0\}.$$

In particular, $\mathrm{rank}(T) = \mathrm{rank}(G) \leq \min(k, n)$.

Proof. This is the standard singular-value correspondence between $AA^\dagger$ and $A^\dagger A$: if $AA^\dagger v = \lambda v$ with $\lambda \neq 0$, then $A^\dagger A(A^\dagger v) = \lambda(A^\dagger v)$ and $A^\dagger v \neq 0$. $\square$

B. Spectral constraints

Proposition 27 (Trace identities)**.** *The structure operator satisfies*

$$\mathrm{tr}(T) = k, \qquad \mathrm{tr}(T^2) = \sum_{i,j=1}^{k} |\langle a_i, a_j\rangle|^2.$$

The diagonal terms $i=j$ each contribute 1, giving $\mathrm{tr}(T^2) \geq k$, with equality if and only if the vectors are mutually orthogonal.

Proof. $\mathrm{tr}(T) = \sum_i \mathrm{tr}(P_{a_i}) = k$. For the second identity, $\mathrm{tr}(T^2) = \sum_{i,j} \mathrm{tr}(P_{a_i}P_{a_j}) = \sum_{i,j} |\langle a_i, a_j\rangle|^2$. Diagonal terms contribute k; off-diagonal terms are non-negative, and vanish simultaneously if and only if all off-diagonal inner products are zero. □

Proposition 28 (Non-orthogonality and maximal eigenvalue)**.** *If the vectors (a_i) are not mutually orthogonal, then $\mathrm{tr}(T^2) > k$ and $\lambda_{\max}(T) > 1$. If, in addition, the nonzero spectrum of T is flat on its support of rank r, then $T = (k/r) I_{\mathrm{supp}\,T}$ and necessarily $k > r$.*

Proof. The strict trace inequality is Proposition 27. If all eigenvalues of T were at most 1, then $\mathrm{tr}(T^2) \leq \mathrm{tr}(T) = k$, a contradiction. Thus $\lambda_{\max}(T) > 1$. The final assertion follows by writing a flat nonzero spectrum as $k/r, \ldots, k/r$. □

Proposition 29 (Majorisation constraint)**.** *Suppose $k \leq n$. Let $\lambda(T) = (\lambda_1, \ldots, \lambda_n)$ be the eigenvalues of T in decreasing order (padded with zeros to length n), and let $d = (1, \ldots, 1, 0, \ldots, 0) \in \mathbb{R}^n$ have k ones and $n-k$ zeros. Then $\lambda(T) \succ d$.*

Proof. The diagonal entries of G are $G_{ii} = \|a_i\|^2 = 1$, so the diagonal of G is $(1, \ldots, 1) \in \mathbb{R}^k$. By the Schur–Horn theorem, $\lambda(G) \succ (1, \ldots, 1)$. Since G and T share the same nonzero eigenvalues (Proposition 26), padding $\lambda(G)$ with $n-k$ zeros gives $\lambda(T)$, and $\lambda(T) \succ d$ follows.

Overcomplete case $k > n$. When $k > n$ the Gram matrix G is $k \times k$ with rank at most n; the majorisation is then best stated for the k-dimensional eigenvalue vector of G against $(1, \ldots, 1) \in \mathbb{R}^k$, and the Schur–Horn argument applies to G directly. □

Proposition 30 (Operator bounds)**.** *The structure operator satisfies $0 \leq T \leq kI$. Moreover, if $\mu := \max_{i \neq j} |\langle a_i, a_j\rangle|$, then*

$$\lambda_{\max}(T) \leq 1 + (k-1)\mu.$$

Proof. Positivity $T \geq 0$ is immediate, and $T \leq kI$ since each $P_{a_i} \leq I$. For the eigenvalue bound, apply the Geršgorin circle theorem to G: since $G_{ii} = 1$ and $|G_{ij}| \leq \mu$ for $i \neq j$, every eigenvalue of G lies in a disk centred at 1 with radius at most $(k-1)\mu$. Since the nonzero eigenvalues of T and G coincide, $\lambda_{\max}(T) \leq 1 + (k-1)\mu$. □

C. Commutant and symmetry

Definition 31 (Commutant)**.** The *unitary commutant* of T is $\mathscr{C}(T) := \{U \in U(H) : UT = TU\}$.

Proposition 32 (Structure of the commutant)**.** *Let $T = \sum_\alpha \lambda_\alpha P_\alpha$ be the spectral decomposition of T with distinct eigenvalues λ_α and orthogonal projections P_α of ranks m_α. Then $\mathscr{C}(T) \cong \prod_\alpha U(m_\alpha)$.*

Remark 33. Spectral degeneracies ($m_\alpha > 1$) give rise to commutant-generated continuous families of distinct rank-one decompositions of T (as in tight frames). A simple spectrum removes this particular symmetry source, although residual decomposition moduli may still occur, as the complex $k=2$ example already illustrates. This explains why the tight-frame case (Remark 19) is especially rich in pseudocontexts.

D. Canonical normal form

Proposition 34 (Spectral normal form)**.** *Every structure operator is unitarily equivalent to $\mathrm{diag}(\lambda_1, \ldots, \lambda_r, 0, \ldots, 0)$ with $\lambda_i > 0$, $\sum_{i=1}^r \lambda_i = k$, and $r = \mathrm{rank}(T)$.*

E. Canonical parametrisation of all decompositions

Theorem 35 (Canonical decomposition)**.** *Let T be a positive operator of rank r with spectral decomposition $T = \sum_{i=1}^r \lambda_i |e_i\rangle\langle e_i|$, $\lambda_i > 0$, and set $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_r)$. For a decomposition into k unit vectors, one necessarily has*

$$\sum_{i=1}^r \lambda_i = \mathrm{tr}\,T = k,$$

and the eigenvalue list satisfies the usual finite-frame Schur–Horn majorisation constraints. In particular, the rank condition $k \geq r$ is necessary but not sufficient by itself.

Every rank-one decomposition $T = \sum_{j=1}^k P_{a_j}$ with $\|a_j\| = 1$ corresponds to a matrix $C \in \mathbb{C}^{r \times k}$ satisfying

$$CC^\dagger = \Lambda, \qquad \|c_j\|^2 = 1 \text{ for all } j, \tag{7}$$

where c_j is the j-th column of C, via $a_j = \sum_{i=1}^r c_{ij} e_i$. Conversely, every such C defines a valid decomposition. Thus the displayed equations give the exact canonical fibre over T; the Schur–Horn constraints are precisely the external existence conditions for this fibre to be nonempty.

Proof. Let $T = AA^\dagger$ with columns a_j. In the eigenbasis of T, write $A = UC$ where U is the unitary embedding the eigenbasis of T on its support into H. Then $CC^\dagger = U^\dagger T U\big|_{\mathrm{supp}(T)} = \Lambda$, and $\|c_j\| = \|a_j\| = 1$. The converse is a direct computation. □

Corollary 36 (Unitary freedom)**.** *Among all rank-one decompositions $T = \sum_{j=1}^k |a_j\rangle\langle a_j|$ into k* not necessarily *unit-norm vectors, any two are related by $A \mapsto AU$ for some $U \in U(k)$ (this is the standard $U(k)$-orbit on the affine fibre $\{A : AA^\dagger = T\}$). The unit-norm constraint $\|c_j\| = 1$ in (7) cuts out a generically real-codimension-$k-1$ subvariety of this orbit (one of the k norm equations is redundant because $\mathrm{tr}\,T = k$), and what is relevant for pseudocontexts is the quotient by phases acting on individual columns and by permutations of columns, since rays—not vectors—are the data.*

F. Characterisation of pseudocontexts

Proposition 37 (Operator equivalence criterion)**.** *Two k-tuples $(a_1,\dots,a_k)$ and $(b_1,\dots,b_k)$ of distinct rays form a pseudocontext if and only if their matrices C_A, C_B in the eigenbasis of T both satisfy* (7)*:*

$$C_A C_A^\dagger = C_B C_B^\dagger = \Lambda.$$

Remark 38. Pseudocontexts correspond to distinct points in the space $\{C \in \mathbb{C}^{r\times k} : CC^\dagger = \Lambda, \|c_j\| = 1\}/\sim$, where $\sim$ identifies decompositions that yield the same set of rays.

G. Extremal spectral bounds

Proposition 39 (Rayleigh bounds)**.** *For any density operator ρ,*

$$\lambda_{\min}(T) \le \mathrm{tr}(\rho T) \le \lambda_{\max}(T).$$

Both bounds are tight, achieved by pure states on the corresponding eigenvectors of T.

VIII. TOPOLOGICAL FORCING VS. ALGEBRAIC COINCIDENCE: THE PROBLEM OF "LOOSE ENDS"

The discrepancy between the purely geometric validity of the $k = 2$ complex pseudocontext and the failure of its orthogonality hypergraph to enforce it highlights a fundamental distinction in quantum contextuality. We categorize pseudocontexts into two types: those that are *graph-forced* (topologically necessitated by the hypergraph) and those that are *algebraic coincidences* (contingent upon specific vector representations).

A. Graph-forced pseudocontexts

For certain tightly intertwined hypergraphs, the completeness condition $P(I) = 1$ is sufficient to structurally force a pseudocontext identity. The covering technique—as demonstrated in the octagon hypergraph—utilizes this. By summing the completeness conditions of selected contexts, one creates two distinct "coverings" of the hypergraph. If the multiplicities of the vertices in both coverings match perfectly, then any variable appearing in both sums must have an equal probability contribution. In such cases, the equality of the probability sums is a rigid, state-independent logical necessity of the hypergraph, applying equally to all probability assignments, whether classical or quantum.

B. Algebraic coincidences: the $k = 2$ case

The $k = 2$ complex pseudocontext formed by $u_\pm$ and $v_\pm$ (Proposition 9) does not admit this topological forcing. Because u_+ and u_- are not orthogonal, they cannot belong to the same context. To use context-completeness equations, each ray must instead be completed separately to an orthonormal basis by adding auxiliary rays. The resulting auxiliary vertices are loose ends unless the hypergraph contains further contexts forcing their contributions to cancel.

Consequently, classical $\{0,1\}$-valuations can assign truth values that violate the pseudocontext identity. The equality $\sum P(u_\pm) = \sum P(v_\pm)$ holds for these specific quantum observables solely due to their vector representation (the structure operator T being identical for both sets), not due to the logical skeleton of their orthogonalities.

To achieve a structurally forced pseudocontext—where the identity is dictated purely by the graph—we require configurations without such loose ends. This necessitates higher-dimensional hypergraphs where pseudocontextuality is embedded directly into the orthomodular logic, effectively closing the "loose ends" that prevent the covering technique from succeeding in simpler cases.

IX. THE OCTAGON PSEUDOCONTEXT VIA DISTINCT COVERINGS

We can rigorously demonstrate the pseudocontextual nature of a tightened—relative to the previous configuration involving $u_\pm$ and $v_\pm$—octagon hypergraph by employing a proof technique based on mutually distinct "coverings," a method we previously utilized in [3].

The hypergraph consists of 20 vertices and 12 contexts of size three. We label the 16 outer vertices along the perimeter as follows: let $v_0, v_1, \dots, v_7$ be the midpoints of the eight octagon edges, and let $v_{01}, v_{12}, \dots, v_{70}$ be the eight corners. The perimeter is formed by eight straight-line contexts:

$$E_i = \{v_{i-1,i},\ v_i,\ v_{i,i+1}\} \quad \text{for } i = 0,\dots,7 \pmod 8.$$

Notice that adjacent midpoints, such as v_1 and v_2, are not on the same context; rather, their respective perimeter contexts E_1 and E_2 intertwine at the shared corner vertex v_{12}.

The remaining four inner contexts connect opposite midpoints and pass through the four inner vertices a_1, a_2 and b_1, b_2:

$$D_0 = \{v_0, a_1, v_4\}, \quad D_2 = \{v_2, a_2, v_6\},$$
$$D_1 = \{v_1, b_2, v_5\}, \quad D_3 = \{v_3, b_1, v_7\}.$$

We aim to prove that the sums of probabilities on the inner pairs match: $P(a_1) + P(a_2) = P(b_1) + P(b_2)$. To do so, we construct two distinct coverings of the hypergraph.

Covering A focuses on the first pseudocontext pair $\{a_1, a_2\}$, as depicted in Fig. 2(a). We construct it by taking the inner contexts D_0, D_2 along with four alternating perimeter edges:

$$A = \{D_0, D_2, E_1, E_3, E_5, E_7\}.$$

Summing the completeness condition over these six contexts yields a total probability of 6. Let us evaluate the multiplicity of each vertex covered by A:

- The inner vertices a_1, a_2 are covered exactly once.
- The midpoints v_0, v_4 and v_2, v_6 are covered exactly once (by D_0, D_2).
- The remaining midpoints v_1, v_3, v_5, v_7 are covered exactly once (by E_1, E_3, E_5, E_7).
- Each of the eight corners $v_{i,i+1}$ is covered exactly once because the selected alternating perimeter contexts perfectly span all corners without overlapping.

Thus, covering A perfectly partitions all 16 outer vertices, covering each exactly once:

$$P(a_1) + P(a_2) + \sum_{i=0}^{7} P(v_i) + \sum_{\text{corners}} P(v_{i,i+1}) = 6. \tag{8}$$

Covering B focuses on the second pseudocontext pair $\{b_1, b_2\}$, depicted in Fig. 2(b). We construct it using the other two inner contexts D_1, D_3 alongside the complementary set of four alternating perimeter edges:

$$B = \{D_1, D_3, E_0, E_2, E_4, E_6\}.$$

Summing over these six contexts also yields 6. Evaluating the vertex multiplicities reveals:

- The inner vertices b_1, b_2 are covered exactly once.
- The midpoints v_1, v_5 and v_3, v_7 are covered exactly once (by D_1, D_3).
- The remaining midpoints v_0, v_2, v_4, v_6 are covered exactly once (by E_0, E_2, E_4, E_6).
- The selected perimeter contexts E_0, E_2, E_4, E_6 again perfectly span all eight corners exactly once.

Hence, Covering B covers the same set of 16 outer vertices, each exactly once:

$$P(b_1) + P(b_2) + \sum_{i=0}^{7} P(v_i) + \sum_{\text{corners}} P(v_{i,i+1}) = 6. \tag{9}$$

Subtracting (9) from (8), all perimeter terms cancel term by term, because each such vertex appears with the same multiplicity in both coverings. The only uncancelled terms are $P(a_1) + P(a_2)$ and $P(b_1) + P(b_2)$, hence

$$P(a_1) + P(a_2) = P(b_1) + P(b_2).$$

This is the desired state-independent identity.

A. The parameter C and $C \neq 1$ octagon FORs

In the octagon coordinatization—that is, the search for a faithful orthogonal representation [5] of the uniform hypergraph [6] in which vectors assigned to vertices within each hyperedge are required to be mutually orthogonal—it is convenient to parametrize the opening of the inner two-pairs by a single positive parameter C. Since only rays matter, all vectors below are understood up to a nonzero scalar multiple.

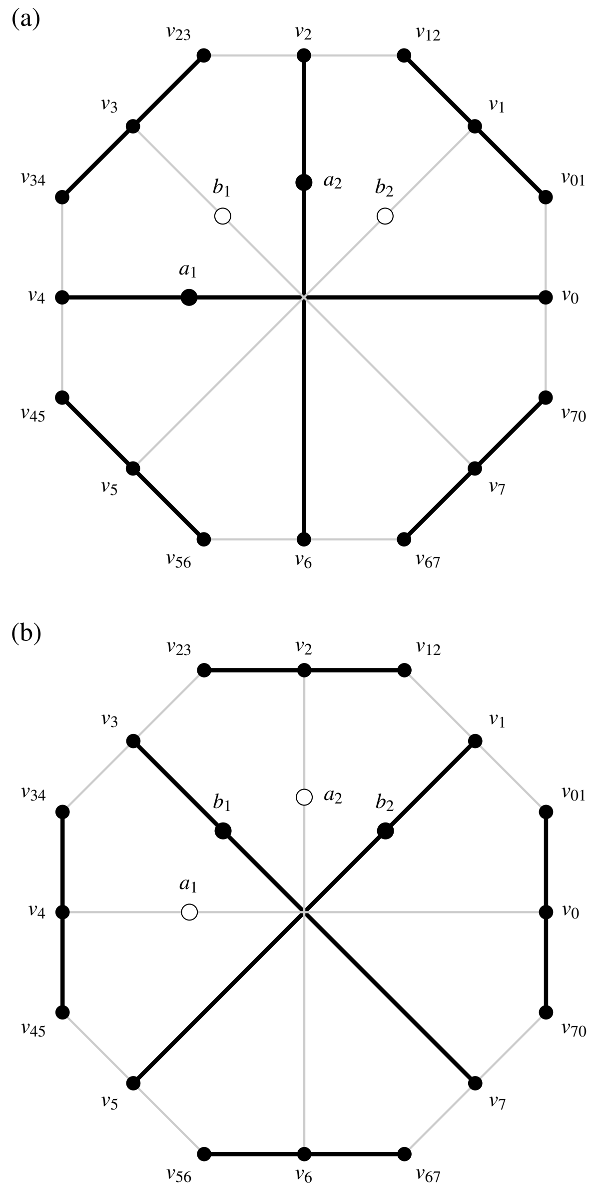

FIG. 2. Graphical representation of the two distinct coverings of the octagon hypergraph. **(a)** Covering A (bold lines) isolates the (a_1, a_2) pseudocontext using inner contexts D_0, D_2 and alternating outer edges. **(b)** Covering B (bold lines) isolates the (b_1, b_2) pseudocontext using inner contexts D_1, D_3 and the complementary outer edges. Vertices actively engaged in a covering are filled black, while the two strictly uncovered inner vertices in each panel are filled white. Since both coverings exactly partition all 16 outer vertices, their respective probability sums must equate, forcing $P(a_1) + P(a_2) = P(b_1) + P(b_2)$.

Proposition 40 (Meaning of the parameter C). *Let $s > 0$, set $C := s^2$, and choose representatives*

$$a_1 = (s, 1, 0), \qquad a_2 = (s, -1, 0),$$

with the companion normal-form pair

$$b_1 = (s, i, 0), \qquad b_2 = (s, -i, 0).$$

Then the inner pair $\{a_1, a_2\}$ has mutual angle 2α satisfying

$$\cos(2\alpha) = \frac{C-1}{C+1}, \qquad \textit{equivalently} \qquad C = \cot^2\alpha.$$

In particular, $C = 1$ is the orthogonal case.

Proof. The normalized representatives of a_1 and a_2 are

$$\frac{(s,1,0)}{\sqrt{s^2+1}}, \qquad \frac{(s,-1,0)}{\sqrt{s^2+1}}.$$

Their inner product equals $(s^2-1)/(s^2+1)$, which gives the stated formula after substituting $C = s^2$. The same computation applies to the b-pair. □

Remark 41. The parameter C measures the nonorthogonality of the inner two-pairs. In the normal form above the companion pair uses the phases i and $-i$. Later, in the explicit $C \neq 1$ example, this normal-form choice is replaced by a phase-generalized pair $(s,\eta,0),(s,-\eta,0)$ with $|\eta| = 1$. This changes the concrete coordinatization but not the common two-projector structure of the inner pseudocontext.

Proposition 42 (Parametrized octagon coordinates)**.** *Let $s > 0$, $C = s^2$, and $N = 1 + C$. Work in the normal form*

$$a_1 = (s,1,0), a_2 = (s,-1,0), b_1 = (s,i,0), b_2 = (s,-i,0).$$

Set

$$\omega = i, \qquad \mu_j = \omega^{2-j}, \qquad j \in \mathbb{Z}/8\mathbb{Z}.$$

Thus

$$(\mu_0,\dots,\mu_7) = (-1,i,1,-i,-1,i,1,-i).$$

Choose $t_0,t_1,t_2,t_3 \in \mathbb{C}^\times$ (the set of non-zero complex numbers), and impose the Hermitian antipodal height relation

$$t_{j+4} = -\frac{N}{\overline{t_j}}, \qquad j = 0,1,2,3.$$

The eight midpoint rays are

$$v_j = (1, s\mu_j, t_j), \qquad j = 0,\dots,7.$$

With this choice the four inner triples

$$\{v_0,a_1,v_4\}, \{v_2,a_2,v_6\}, \{v_1,b_2,v_5\}, \{v_3,b_1,v_7\}$$

are automatically contexts.

Define the Gram kernel of the midpoint rays by

$$G_{jk} := \langle v_j, v_k\rangle = 1 + C\overline{\mu_j}\mu_k + \overline{t_j}t_k.$$

The corner rays are obtained by the Hermitian cross-product convention

$$v_{j,j+1} = \overline{v_j \times v_{j+1}}, \qquad j \in \mathbb{Z}/8\mathbb{Z},$$

where $\times$ is the ordinary bilinear cross product, and the index arithmetic is taken modulo 8 to reflect the closed cyclic perimeter of the octagon. Each such corner is orthogonal to its two adjacent midpoints. Therefore the remaining condition for the eight perimeter triples

$$E_j = \{v_{j-1,j}, v_j, v_{j,j+1}\}, \qquad j \in \mathbb{Z}/8\mathbb{Z},$$

to be contexts is equivalently the cyclic Gram system

$$G_{j-1,j}G_{j,j+1} = G_{j-1,j+1}G_{j,j}, \qquad j \in \mathbb{Z}/8\mathbb{Z}. \tag{10}$$

In expanded midpoint form this is

$$\begin{aligned}&\left(1 + C\overline{\mu_{j-1}}\mu_j + \overline{t_{j-1}}t_j\right)\left(1 + C\overline{\mu_j}\mu_{j+1} + \overline{t_j}t_{j+1}\right)\\ &\qquad - \left(1 + C\overline{\mu_{j-1}}\mu_{j+1} + \overline{t_{j-1}}t_{j+1}\right)\left(N + |t_j|^2\right) = 0. \end{aligned}\tag{11}$$

Thus any choice of t_0,t_1,t_2,t_3 satisfying these closure equations yields an orthogonal representation of the octagon hypergraph. It is faithful exactly when the listed rays are pairwise distinct as vertices and no orthogonalities occur except those prescribed by the twelve contexts.

Proof. The antipodal height relation gives, for $j = 0,1,2,3$,

$$\langle v_j, v_{j+4}\rangle = 1 + C + \overline{t_j}t_{j+4} = N - N = 0.$$

Moreover, the displayed values of μ_j make v_0, v_4 orthogonal to a_1, v_2, v_6 orthogonal to a_2, v_1, v_5 orthogonal to b_2, and v_3, v_7 orthogonal to b_1. Hence the four inner triples are contexts.

For the perimeter, the vector $\overline{x \times y}$ is orthogonal to both x and y with respect to the Hermitian inner product. Indeed, for

$$v_j = (1, s\omega^{2-j}, t_j),$$

a direct computation gives

$$\overline{v_j \times v_{j+1}} = \left(s\omega^{j-2}\left(\overline{t_{j+1}} - \omega\overline{t_j}\right), \overline{t_j} - \overline{t_{j+1}}, s\omega^{j-2}(\omega - 1)\right),$$

or, up to a nonzero scalar,

$$v_{j,j+1} \propto \left(s\left(\overline{t_{j+1}} - \omega\overline{t_j}\right), \omega^{2-j}\left(\overline{t_j} - \overline{t_{j+1}}\right), s(\omega-1)\right).$$

Thus $v_{j,j+1}$ is orthogonal to v_j and v_{j+1}. The only remaining orthogonality in the perimeter context E_j is the orthogonality of the two adjacent corners. The standard Gram-determinant identity for the two cross products gives

$$\langle v_{j-1}, v_j\rangle\langle v_j, v_{j+1}\rangle - \langle v_{j-1}, v_{j+1}\rangle\|v_j\|^2 = 0,$$

which is precisely (10); expanding the entries of G gives (11). □

Example 43 (A faithful $C \neq 1$ pseudocontext FOR)**.** Set

$$C = \frac{49 - 5\sqrt{69}}{26}, \qquad s = \sqrt{C}, \qquad N = 1 + C, \qquad r = \sqrt{N}.$$

Then $C > 0$ and $C \neq 1$; indeed C is the smaller positive root of

$$13C^2 - 49C + 13 = 0.$$

Let

$$\eta = \frac{5 + 12i}{13}$$

and

$$\zeta=\frac{14+6\sqrt{69}}{65}+i\,\frac{-21+4\sqrt{69}}{65}.$$

Both η and ζ have unit modulus.

Define the two inner pairs

$$a_1=(s,1,0),\qquad a_2=(s,-1,0),$$

and

$$b_1=(s,\eta,0),\qquad b_2=(s,-\eta,0).$$

The pairs $\{a_1,a_2\}$ and $\{b_1,b_2\}$ are nonorthogonal because

$$\langle a_1,a_2\rangle=\langle b_1,b_2\rangle=C-1\neq 0.$$

Moreover, since $\eta\notin\{\pm1\}$, the four rays a_1,a_2,b_1,b_2 are pairwise distinct. Since $|\eta|=1$, their normalized projector sums agree:

$$P_{a_1}+P_{a_2}=P_{b_1}+P_{b_2}.$$

Thus the two inner pairs form a genuine complex pseudocontext.

Now put

$$(\mu_0,\dots,\mu_3)=(-1,\eta,1,-\eta),\,(\mu_4,\dots,\mu_7)=(-1,\eta,1,-\eta),$$
$$(\tau_0,\dots,\tau_3)=(1,\zeta,-i,-i\zeta),\,(\tau_4,\dots,\tau_7)=(-1,-\zeta,i,i\zeta),$$

and define the eight midpoint rays by

$$v_j=(1,s\mu_j,r\tau_j),\qquad j=0,\dots,7.$$

The corner rays are

$$v_{j,j+1}=\overline{v_j\times v_{j+1}},\qquad j\in\mathbb{Z}/8\mathbb{Z},$$

where $\times$ is the ordinary bilinear cross product.

The twelve contexts are the eight perimeter contexts

$$\{v_{j-1,j},v_j,v_{j,j+1}\},\qquad j\in\mathbb{Z}/8\mathbb{Z},$$

together with the four inner contexts

$$\{v_0,a_1,v_4\},\qquad \{v_2,a_2,v_6\},$$
$$\{v_1,b_2,v_5\},\qquad \{v_3,b_1,v_7\}.$$

Together with the faithfulness check in the following remark, these coordinates give a faithful orthogonal representation of the octagon hypergraph carrying the displayed $C\neq 1$ pseudocontext.

Remark 44 (Reduction of the symmetric ansatz). For the phase-generalized symmetric ansatz in Example 43, the eight perimeter closure equations become identical once the unit-phase condition $|\zeta|=1$ is imposed. Before imposing this condition, the alternating equations differ only by a multiple of $|\zeta|^2-1$. This is best seen before substituting the displayed value of C. Keep

$$\eta=\frac{5+12i}{13},\qquad \zeta=x+iy,\qquad x,y\in\mathbb{R},$$

and impose the same cyclic pattern

$$(\mu_0,\dots,\mu_3)=(-1,\eta,1,-\eta),\,(\mu_4,\dots,\mu_7)=(-1,\eta,1,-\eta),$$
$$(\tau_0,\dots,\tau_3)=(1,\zeta,-i,-i\zeta),\,(\tau_4,\dots,\tau_7)=(-1,-\zeta,i,i\zeta),$$

with midpoint heights $t_j=r\tau_j$, $r^2=N=1+C$. Substitution in (11) gives the same equation for every $j\in\mathbb{Z}/8\mathbb{Z}$. Written in real and imaginary parts, this single complex equation is equivalent to

$$7Cx-17Cy-13C-13x+13y+13=0$$

and $-7C^2x+17C^2y+13C^2-20Cx+30Cy+2C-13x+13y+13=0$. Solving these two linear equations for x and y gives

$$x=\frac{104-76C}{65(1+C)},\qquad y=\frac{39-81C}{65(1+C)}.$$

The unit-circle condition $|\zeta|=1$, i.e. $x^2+y^2=1$, then reduces to

$$\frac{48\left(13C^2-49C+13\right)}{325(1+C)^2}=0.$$

Thus the symmetric ansatz closes precisely when

$$13C^2-49C+13=0.$$

For the smaller positive root $C=(49-5\sqrt{69})/26$, the above formulas for x and y give exactly

$$\zeta=\frac{14+6\sqrt{69}}{65}+i\,\frac{-21+4\sqrt{69}}{65},$$

as used in the example.

Remark 45 (Independent numerical verification). For the coordinates in Example 43, a direct Hermitian inner-product check on the full set of twenty rays gives

$$\max_{\{x,y\}\text{ prescribed}}\frac{|\langle x,y\rangle|}{\|x\|\,\|y\|}<1.4\times10^{-16}.$$

There are 36 prescribed orthogonal pairs among the twelve contexts, and all vanish to this tolerance. Among the remaining 154 unprescribed pairs, the smallest normalized inner product is

$$0.2=\frac{1}{5},$$

so no unintended orthogonality occurs. All twenty ray representatives are nonzero and projectively distinct; the smallest projective separation between distinct listed rays is

$$0.4472135955=\frac{1}{\sqrt{5}}.$$

Finally,

$$\left\|P_{a_1}+P_{a_2}-P_{b_1}-P_{b_2}\right\|_{\mathrm{F}}<2.8\times10^{-16}.$$

Thus the orthogonalities prescribed by the twelve contexts hold to machine precision, no additional orthogonalities occur, and the pseudocontext identity of the inner two-pairs is verified numerically as well as algebraically.

B. Exact $\mathbb{C}^3$ coordinates at $C = 1$

At $C = 1$ the inner pair becomes orthogonal, and the numerical $\mathbb{C}^3$ solution found for the octagon admits a closed-form algebraic realization. As in the main text, the vectors below are ray representatives; normalization is not essential. The configuration is depicted in Fig. 3. A closer inspection in Appendix B 2 shows that this configuration of 21 points in 14 contexts (hyperedges) supports a separating—and thus set representable [7], as discussed in Chapter 3 of Ref. [8]; see Theorem 0 of Ref. [9]—set of 24 two-valued states.

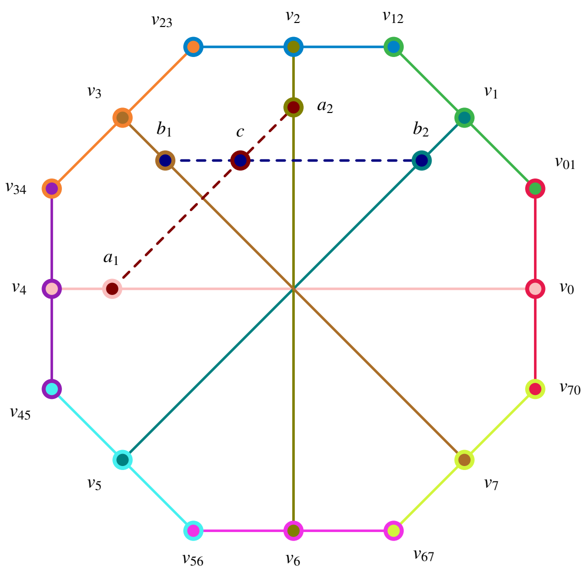

FIG. 3. Redraw of the octagon using exactly the same coordinates as Fig. 2(a)/(b). The 12 original contexts are drawn with solid lines in 12 distinct colors. The two additional $C = 1$ contexts are dashed and meet at the intertwining vertex c. Concentric disks mark vertices incident with two contexts.

Example 46 (Exact $\mathbb{C}^3$ realization at $C = 1$)**.** Let

$$p = \frac{\sqrt{14}+\sqrt{2}}{4}, \qquad q = \frac{\sqrt{14}-\sqrt{2}}{4}.$$

Then

$$p^2 + q^2 = 2, \qquad pq = \frac{3}{4}.$$

A convenient choice of the inner rays intersection at the additional point $c = (0,0,1)$ is

$$a_1 = (1,1,0), \quad a_2 = (1,-1,0),$$
$$b_1 = (1,i,0), \quad b_2 = (1,-i,0).$$

One compatible choice of the eight midpoint rays is

$$v_0 = (1,-1,\sqrt{2}), \qquad v_4 = (1,-1,-\sqrt{2}),$$
$$v_1 = (1,i,p+iq), \qquad v_5 = (1,i,-p-iq),$$
$$v_2 = (1,1,-i\sqrt{2}), \qquad v_6 = (1,1,i\sqrt{2}),$$
$$v_3 = (1,-i,q-ip), \qquad v_7 = (1,-i,-q+ip).$$

One compatible choice of the eight corner rays is

$$v_{01} = \big(-p+i(q+\sqrt{2}),\ \sqrt{2}-p+iq,\ 1-i\big),$$
$$v_{12} = \big(\sqrt{2}-p+iq,\ p-i(q+\sqrt{2}),\ 1+i\big),$$
$$v_{23} = \big(q+\sqrt{2}+ip,\ -q+i(\sqrt{2}-p),\ -1+i\big),$$
$$v_{34} = \big(q+i(p-\sqrt{2}),\ q+\sqrt{2}+ip,\ -1-i\big),$$
$$v_{45} = \big(p-i(q+\sqrt{2}),\ -\sqrt{2}+p-iq,\ 1-i\big),$$
$$v_{56} = \big(-\sqrt{2}+p-iq,\ -p+i(q+\sqrt{2}),\ 1+i\big),$$
$$v_{67} = \big(-q-\sqrt{2}-ip,\ q-i(\sqrt{2}-p),\ -1+i\big),$$
$$v_{70} = \big(-q-i(p-\sqrt{2}),\ -q-\sqrt{2}-ip,\ -1-i\big).$$

A direct symbolic check shows that each listed context is orthogonal. The identities $p^2 + q^2 = 2$ and $pq = \frac{3}{4}$ are sufficient for the simplification. Hence the octagon from Section IX admits an exact $\mathbb{C}^3$ realization at $C = 1$.

Remark 47. The coordinates above are not unique. Any common unitary transformation, together with arbitrary rescalings of the individual ray representatives, yields an equivalent realization.

Remark 48 (Phase symmetry). The exact $\mathbb{C}^3$ realization exploits a complex phase symmetry in the third coordinate. This is one of the reasons why the construction is much more rigid over $\mathbb{R}$ than over $\mathbb{C}$.

C. No faithful real realization at $C = 1$

Example 46 provides an exact realization of the octagon hypergraph in $\mathbb{C}^3$ at the orthogonal limit $C = 1$, where the inner pseudocontext splits into two contexts. We show that no faithful realization by real rays exists.

Proposition 49 (No faithful real realization at $C = 1$)**.** *The octagon hypergraph of Section IX, augmented by the two inner $C = 1$ contexts intersecting at the vertex c, admits no faithful representation by rays in $\mathbb{R}^3$.*

Proof. Assume a faithful real realization exists. By a global orthogonal transformation, fix $c = (0,0,1)^T$. All rays sharing a context with c lie in the xy-plane. Using rotational freedom in that plane, set

$$a_1 = (1,0,0)^T, \qquad a_2 = (0,1,0)^T.$$

Let

$$b_1 = (\cos\theta, \sin\theta, 0)^T, \qquad b_2 = (-\sin\theta, \cos\theta, 0)^T,$$

with $\theta \notin \frac{\pi}{2}\mathbb{Z}$ by faithfulness. Define $m = \tan\theta \neq 0$ and $M = 1 + m^2 > 1$.

Let the midpoint rays be $v_j = (X_j, Y_j, Z_j)^T$.

Nonplanarity of midpoint rays. Every midpoint lies in exactly one inner context, paired with one of the planar reference vectors: $v_0, v_4 \in D_0$ (with a_1); $v_2, v_6 \in D_2$ (with a_2); $v_1, v_5 \in D_1$ (with b_2); $v_3, v_7 \in D_3$ (with b_1). Suppose $Z_j = 0$, so that v_j lies in the xy-plane. If v_j is paired with a_1 (resp. a_2), orthogonality forces $X_j = 0$ (resp. $Y_j = 0$), leaving v_j proportional to a_2 (resp. a_1). If v_j is paired with b_1 or b_2, the relations $Y = mX$ or $X = -mY$ together with $Z_j = 0$ force coincidence with one of the b-rays. In every case this violates faithfulness. Hence $Z_j \neq 0$ for all j, and we may normalize

$$v_j = (X_j, Y_j, 1)^T.$$

The inner contexts impose:

$$\begin{aligned} D_0: &\quad X_0 = X_4 = 0, \qquad Y_0 Y_4 + 1 = 0, \\ D_1: &\quad Y_1 = mX_1, \quad Y_5 = mX_5, \quad X_1 X_5 M + 1 = 0, \\ D_3: &\quad X_3 = -mY_3, \quad X_7 = -mY_7, \quad Y_3 Y_7 M + 1 = 0. \end{aligned}$$

Thus

$$X_5 = -\frac{1}{MX_1}, \qquad Y_7 = -\frac{1}{MY_3}.$$

The quantities X_1 and Y_3 are nonzero: if $X_1 = 0$ (resp. $Y_3 = 0$), then v_1 (resp. v_3) becomes collinear with a coordinate direction, forcing vertex identifications incompatible with faithfulness.

For perimeter contexts, the condition that adjacent corner rays are orthogonal yields

$$(v_{j-1} \cdot v_j)(v_j \cdot v_{j+1}) - (v_{j-1} \cdot v_{j+1}) \|v_j\|^2 = 0.$$

Evaluating at $j = 0$ and $j = 4$ gives

$$Y_0 = \frac{Y_7 + mX_1}{1 - mX_1 Y_7}, \qquad Y_4 = \frac{Y_3 + mX_5}{1 - mY_3 X_5}, \tag{12}$$

where the denominators are nonvanishing: if $1 - mX_1 Y_7 = 0$, the closure equation at $j = 0$ reduces to $Y_7 + mX_1 = 0$, and combining this with $mX_1 Y_7 = 1$ yields $-m^2 X_1^2 = 1$, which has no real solution. The case $1 - mY_3 X_5 = 0$ at $j = 4$ is excluded analogously.

Substituting the inner relations into (12) and simplifying the compound fractions yields

$$Y_0 = \frac{mMX_1 Y_3 - 1}{MY_3 + mX_1}, \qquad Y_4 = \frac{MX_1 Y_3 - m}{MX_1 + mY_3}.$$

Using $Y_0 Y_4 = -1$ from D_0, set $u = X_1 Y_3$. Expanding the product of these fractions, collecting terms in u, and applying the identity $M = 1 + m^2$ causes all mixed terms to cancel, yielding the master constraint (see Appendix A)

$$(M^2 u^2 + mu + 1) + M(X_1^2 + Y_3^2) = 0. \tag{13}$$

Since $X_1, Y_3 \in \mathbb{R}$,

$$X_1^2 + Y_3^2 \geq 2|X_1 Y_3| = 2|u|,$$

and therefore the left-hand side is bounded below by

$$f(u) = M^2 u^2 + mu + 1 + 2M|u|.$$

Using $M = 1 + m^2$, one has $2M - |m| > 0$, hence

$$M^2 u^2 + mu + 2M|u| \geq M^2 u^2 + |u|(2M - |m|) \geq 0,$$

with equality only at $u = 0$. Consequently $f(u) \geq 1$ for all $u \in \mathbb{R}$, contradicting the required vanishing. Therefore no real realization exists. □

Remark 50 (Minimality of the obstruction). Of the fourteen orthogonality relations defining the $C = 1$ octagon, the proof above invokes only the three inner contexts D_0, D_1, D_3 and the two perimeter closures at $j = 0$ and $j = 4$; the remaining nine relations are not used. The impossibility therefore holds already for this five-relation sub-pattern.

Remark 51 (Complex realization). In the complex construction, one has formally $m = i$, hence $M = 1 + i^2 = 0$. The relations above then become singular, and the coercive quadratic structure responsible for the contradiction disappears. Substituting $M = 0$ reduces the constraint to $iu + 1 = 0$, which admits solutions in $\mathbb{C}$.

Remark 52. The paper [10] presented the first construction of an orthogonality (Greechie) diagram which can be represented by vectors in complex spaces, but not in real ones. As a by-product of our effort, we found another (and much simpler) such diagram described in this section.

X. PHYSICAL INTERPRETATION AND CONSTRAINTS

A pseudocontext $(a_1, \ldots, a_k)$, $(b_1, \ldots, b_k)$ implements a state-independent empirical equivalence between two families of rank-one observables $A_i = P_{a_i}$ and $B_i = P_{b_i}$. Although the operators within each family are generally noncommuting, the two linear functionals

$$\rho \longmapsto \sum_i \mathrm{tr}(\rho A_i), \qquad \rho \longmapsto \sum_i \mathrm{tr}(\rho B_i)$$

agree on the whole state space.

The spectral decomposition of T establishes absolute quantum limits on these probability sums. By Proposition 39,

$$\lambda_{\min}(T) \leq \sum_{i=1}^{k} P(a_i) \leq \lambda_{\max}(T). \tag{14}$$

These bounds are tight.

For the 36-atom hypergraph example [4], the structure operator $T_2 = \mathrm{diag}(1/5, 1/5, 13/5)$ confines the quantum probability sum to $[1/5, 13/5] = [0.2, 2.6]$. A classical hidden-variable model based on two-valued ($\{0,1\}$) truth assignments to projections—subject to the consistency requirement that exactly one atom in each orthogonal basis receives the value 1—would allow the sum to range over $\{0,1,2,3\}$, which includes the extreme values 0 and 3. Because T_2 has no eigenvalue equal to 0 or 3, quantum mechanics forbids these extremes; the quantum system cannot reach them. This discrepancy is a signature of Kochen–Specker-type contextuality [9, 11].

This state-independent additive equivalence is related to, but distinct from, the operational notion of contextuality introduced by Spekkens [12], which applies to arbitrary experimental procedures (preparations, transformations, and unsharp measurements) rather than to sharp projective measurements alone. A precise comparison is beyond the scope of this note.

In the language of quantum information, $(A_1, \ldots, A_k)$ and $(B_1, \ldots, B_k)$ are distinct rank-one decompositions of the same positive operator. When this operator is proportional to the identity on its support, the families form tight frames; after normalisation they yield POVMs.

XI. GENUINELY THREE-DIMENSIONAL REAL PSEUDOCONTEXTS

The constructions of Sections V–VI produce real pseudocontexts whose rays are confined to a two-dimensional subspace. We now exhibit the full-support tight-frame mechanism listed in the final row of Table I. It exploits the spectral structure developed in Section VII and yields continuous families of real pseudocontexts whose rays span all of $\mathbb{R}^n$. The starting point is the observation that when the structure operator T is proportional to the identity, the unitary commutant (Proposition 32) is the full orthogonal group, so every spatial rotation produces a new rank-one decomposition of the same operator.

A. The rotation principle for tight frames

Proposition 53 (Rotation principle)**.** *Let $(a_1,\dots,a_k)$ be a unit-norm tight frame in $\mathbb{R}^n$ with pairwise distinct rays, i.e.,*

$$T=\sum_{i=1}^{k}P_{a_i}=\frac{k}{n}I_n.$$

For every $R\in O(n)$, the rotated tuple $(Ra_1,\dots,Ra_k)$ is again a unit-norm tight frame with the same structure operator T. The collision set

$$S^{*}=\left\{R\in O(n):[Ra_j]=[a_i]\text{ for some }i,j\right\}$$

is a closed proper subvariety of $O(n)$ of codimension at least $n-1$, and hence has Haar measure zero. For every $R\in O(n)\setminus S^{*}$ the $2k$ rays $[a_1],\dots,[a_k],[Ra_1],\dots,[Ra_k]$ are pairwise distinct, and the tuples $(a_1,\dots,a_k)$ and $(Ra_1,\dots,Ra_k)$ form a pseudocontext of size k, genuine whenever $k>n$.

Proof. Conjugation by $R\in O(n)$ fixes any scalar multiple of the identity, so $RTR^{T}=T$. Since $P_{Ra_i}=RP_{a_i}R^{T}$, summing gives $\sum_i P_{Ra_i}=RTR^{T}=T$, proving that the rotated tuple is a unit-norm tight frame with the same structure operator.

For each ordered pair (i,j) and each sign $\varepsilon\in\{\pm1\}$, the locus $\{R\in O(n):Ra_j=\varepsilon a_i\}$ is a single coset of the stabiliser $\mathrm{Stab}_{O(n)}(a_j)\cong O(n-1)$, hence a closed submanifold of codimension $n-1$ in $O(n)$. The collision set S^{*} is the finite union of these submanifolds, hence a closed subvariety of codimension at least $n-1$ and Haar measure zero.

For $R\in O(n)\setminus S^{*}$, no Ra_j is collinear with any a_i. Since the original rays are pairwise distinct and rotations preserve ray distinctness, the $2k$ rays are pairwise distinct. When $k>n$, no k-tuple of unit vectors in $\mathbb{R}^n$ can be orthogonal, so neither tuple is a context, and the pseudocontext is genuine. □

Remark 54. The genericity condition $R\in O(n)\setminus S^{*}$ is mild: S^{*} has Haar measure zero in $O(n)$, so a Haar-random rotation produces a pseudocontext with probability one. The valid parameter space $O(n)\setminus S^{*}$ is open and dense, and itself a smooth manifold of dimension $\binom{n}{2}$.

Remark 55 (Spectral interpretation). Proposition 53 is the maximally degenerate case of the unitary commutant analysis of Section VII: when the spectrum of T collapses to a single eigenvalue, the commutant $\mathscr{C}(T)$ becomes the full orthogonal group, and every rotation gives a fresh rank-one decomposition. This sharpens the informal observation made in Remark 19 for the two-dimensional support of the trine: tight frames are precisely the configurations whose pseudocontextual partners form a continuous orbit under a continuous Lie group of symmetries.

Corollary 56 (Spectrum of tight pseudocontexts)**.** *Let $n\geq 2$. A genuine pseudocontext with structure operator $T=\lambda I_n$ exists if and only if λ is of the form*

$$\lambda=\frac{k}{n}$$

for some integer $k>n$.

Proof. If $T=\lambda I_n$ arises from a k-tuple of unit vectors, then by Proposition 27,

$$\mathrm{tr}(T)=k.$$

Since $\mathrm{tr}(\lambda I_n)=n\lambda$, it follows that $\lambda=k/n$.

Moreover, $\mathrm{rank}(T)=n$, while $\mathrm{rank}(T)\leq k$, hence $k\geq n$. If $k=n$, then Proposition 27 implies that the vectors are mutually orthogonal, so the tuple is a context rather than a genuine pseudocontext. Thus $k>n$ is necessary.

Conversely, for every integer $k>n$, the standard finite unit-norm tight-frame existence theorem supplies a frame $(a_1,\dots,a_k)$ with pairwise distinct rays in $\mathbb{R}^n$ (and hence in $\mathbb{C}^n$) satisfying

$$\sum_{i=1}^{k}P_{a_i}=\frac{k}{n}I_n.$$

Proposition 53 then implies that, for a generic rotation $R\in O(n)\setminus S^{*}$, the tuple $(Ra_1,\dots,Ra_k)$ yields a distinct decomposition of the same operator. The two tuples therefore form a genuine pseudocontext. □

Remark 57 (Discrete spectrum of tight operators). The scalar λ is constrained to the discrete set

$$\lambda\in\left\{\frac{k}{n}:k\in\mathbb{N},\ k>n\right\}.$$

In particular, not every positive real value can occur. For fixed dimension n, the admissible values form a discrete subset of $(1,\infty)$; as n varies, these values become dense in $(1,\infty)$.

B. The tetrahedral pseudocontext

The simplest nontrivial application of Proposition 53 in genuine three-dimensional ambient space is provided by the regular tetrahedron, the unique (up to orthogonal equivalence and relabeling) unit-norm tight frame of size four in $\mathbb{R}^3$.

Example 58 (Tetrahedral pseudocontext)**.** Let

$$\begin{aligned} a_1 &= (0, 0, 1),\\ a_2 &= \left(\tfrac{2\sqrt{2}}{3}, 0, -\tfrac{1}{3}\right),\\ a_3 &= \left(-\tfrac{\sqrt{2}}{3}, \tfrac{\sqrt{6}}{3}, -\tfrac{1}{3}\right),\\ a_4 &= \left(-\tfrac{\sqrt{2}}{3}, -\tfrac{\sqrt{6}}{3}, -\tfrac{1}{3}\right) \end{aligned}$$

be the vertices of a regular tetrahedron inscribed in the unit sphere of $\mathbb{R}^3$, satisfying $\langle a_i, a_j\rangle = -1/3$ for $i \neq j$. Choose any $R \in SO(3) \setminus S^*$ (i.e. R is not a rotation that maps any a_j onto an $\pm a_i$), and define

$$b_j := Ra_j, \qquad j = 1, \ldots, 4.$$

Then the eight rays $[a_1], \ldots, [a_4], [b_1], \ldots, [b_4]$ are pairwise distinct, and

$$\sum_{j=1}^{4} P_{a_j} = \sum_{j=1}^{4} P_{b_j} = \tfrac{4}{3} I_3,$$

so $(a_1, \ldots, a_4)$ and $(b_1, \ldots, b_4)$ form a genuine pseudocontext of size $k = 4$ in $\mathbb{R}^3$ whose eight rays span all of $\mathbb{R}^3$.

A concrete instance is provided by the quarter-turn about the x-axis,

$$R = \begin{pmatrix} 1 & 0 & 0\\ 0 & 0 & -1\\ 0 & 1 & 0 \end{pmatrix} \in SO(3),$$

which yields

$$\begin{aligned} b_1 &= (0, -1, 0),\\ b_2 &= \left(\tfrac{2\sqrt{2}}{3}, \tfrac{1}{3}, 0\right),\\ b_3 &= \left(-\tfrac{\sqrt{2}}{3}, \tfrac{1}{3}, \tfrac{\sqrt{6}}{3}\right),\\ b_4 &= \left(-\tfrac{\sqrt{2}}{3}, \tfrac{1}{3}, -\tfrac{\sqrt{6}}{3}\right). \end{aligned}$$

A direct check shows that none of the b_j is collinear with any a_i, confirming that the eight rays are distinct.

Proof. The matrix $A = [a_1\ a_2\ a_3\ a_4]$ satisfies $AA^T = \frac{4}{3} I_3$ by direct computation, so $T = \frac{4}{3} I_3$. Since $R \in O(3)$,

$$\sum_{j=1}^{4} P_{b_j} = R\Big(\sum_{j=1}^{4} P_{a_j}\Big)R^T = R \cdot \tfrac{4}{3} I_3 \cdot R^T = \tfrac{4}{3} I_3,$$

giving the common structure operator. The hypothesis $R \in SO(3) \setminus S^*$ excludes ray collisions, so the eight rays are pairwise distinct. Within the a-tuple, the inner products $\langle a_i, a_j\rangle = -1/3 \neq 0$ for $i \neq j$ show that no two distinct vectors are orthogonal, so the tuple is not a context; the b-tuple is its image under the orthogonal map R and inherits this property. The pseudocontext is therefore genuine, in agreement with Proposition 53 specialised to $n = 3$, $k = 4$. □

Remark 59 (Orbit of a fixed tetrahedron). For the fixed tetrahedral frame in Example 58, the family obtained by varying $R \in SO(3) \setminus S^*$ is parametrized by an open dense subset of $SO(3)$, modulo the finite tetrahedral symmetry group. This is a 3-dimensional orbit of one fixed frame, not the full moduli space of all size-4 tight frames in $\mathbb{R}^3$.

Remark 60 (Fixed-frame versus full moduli). The tetrahedral example exhibits one particularly symmetric point in the space of size-4 tight frames in $\mathbb{R}^3$. The point of the example is only that this one orbit already yields a genuine pseudocontext spanning all of $\mathbb{R}^3$.

Remark 61 (Physical interpretation). A four-outcome rank-one POVM in $\mathbb{R}^3$ structured like a tetrahedral measurement apparatus may be rotated rigidly by any $R \in SO(3)$. The rotated apparatus describes a physically distinct set of nonorthogonal measurement directions, yet by Example 58 the sum of single-outcome probabilities coincides on every quantum state. The pseudocontextual identity at $k = 4$ is therefore not an isolated algebraic accident but a robust feature of the tetrahedral frame geometry.

Conjecture 62 (The tetrahedral pseudocontext is not graph–forced)**.** *Let $(a_1, \ldots, a_4)$ and $(b_1, \ldots, b_4) = (Ra_1, \ldots, Ra_4)$ be the tetrahedral pseudocontext of Example 58, with $R \in SO(3) \setminus S^*$. Then the pseudocontextual identity*

$$\sum_{j=1}^{4} P_{a_j} = \sum_{j=1}^{4} P_{b_j} = \tfrac{4}{3} I_3$$

is not graph-forced over $\mathbb{R}$: there is no finite orthogonality hypergraph Γ over $\mathbb{R}^3$ with vertex set containing $\{[a_1], \ldots, [a_4], [b_1], \ldots, [b_4]\}$ such that the closure conditions $\sum_{v \in C} P(v) = 1$ for C ranging over the contexts of Γ entail $\sum_j P(a_j) = \sum_j P(b_j)$ via a covering argument of the type employed in Section IX, for any choice of $R \in SO(3) \setminus S^*$ outside a measure-zero subset.

Remark 63 (Supporting evidence). The eight rays $[a_1], \ldots, [a_4], [b_1], \ldots, [b_4]$ carry no pairwise orthogonalities among themselves: within each tuple $\langle a_i, a_j\rangle = \langle b_i, b_j\rangle = -\frac{1}{3} \neq 0$ for $i \neq j$, and the cross-orthogonality locus $\{R \in SO(3) : \langle a_i, Ra_j\rangle = 0 \text{ for some } i, j\}$ is a positive-codimension subvariety of $SO(3)$. The minimal hypergraph extension that could cancel both tuples on a shared set of auxiliaries requires eight extra unit vectors $w_k \propto a_{i(k)} \times b_{j(k)}$ subject to eight cross-Gram product conditions of the form

$$\langle a_i, b_{j_1}\rangle \langle a_i, b_{j_2}\rangle \;=\; -\tfrac{1}{3},$$

arising from mutual orthogonality of the auxiliaries inside each context. These conditions are eight algebraic equations on the nine entries of the cross-Gram matrix $G = A^T RA$, while $R \in SO(3)$ contributes only three parameters; generically the system is inconsistent, and the special loci where it is consistent collapse onto S^* itself, violating distinctness. This argument rules out the minimal embedding; the conjecture asserts that the same rigidity persists for arbitrarily complex extensions.

Remark 64 (Complex analogue). Over $\mathbb{C}$ the parameter count flips: a rotation $U \in U(3)$ has nine real parameters while the cross-Gram entries $\langle a_i, Ub_j\rangle$ are complex. The dimension argument above does not apply, leaving open whether complex tetrahedral-type pseudocontexts admit hypergraph-forcing extensions in $\mathbb{C}^3$.

C. The tightness obstruction at $k = 3$

The rotation principle of Proposition 53 cannot be applied at $k = 3$ in $\mathbb{R}^3$, because tightness in this regime collapses to orthogonality.

Proposition 65. *Let (a_1, a_2, a_3) be a unit-norm tight frame in $\mathbb{R}^3$, i.e. $T = I_3$. Then (a_1, a_2, a_3) is an orthonormal basis, hence a context.*

Proof. By Proposition 27, $\mathrm{tr}(T^2) = k + \sum_{i \neq j} |\langle a_i, a_j \rangle|^2$. With $T = I_3$ and $k = 3$ this gives $3 = 3 + \sum_{i \neq j} |\langle a_i, a_j \rangle|^2$, forcing all off-diagonal Gram entries to vanish. ☐

Corollary 66. *Every genuine real pseudocontext of size $k = 3$ whose rays span all of $\mathbb{R}^3$ has nondegenerate structure operator, and therefore lies outside the rotational orbit construction of Proposition 53.*

Such genuinely three-dimensional, nondegenerate $k = 3$ pseudocontexts do exist: explicit coordinatisations were obtained in [4] (recalled in Remark 23), with structure operators having spectra $\{2, (7 + \sqrt{21})/14, (7 - \sqrt{21})/14\}$ and $\{1/5, 1/5, 13/5\}$ respectively. These configurations are produced not by acting on a single tight base configuration with a continuous symmetry, but by solving the canonical fibre conditions of Theorem 35 directly with a prescribed nondegenerate spectrum, the off-diagonal Gram data being constrained by an underlying orthogonality hypergraph rather than recoverable from a continuous group orbit.

D. Larger sizes

For every $k \geq 4$, unit-norm tight frames in $\mathbb{R}^3$ exist in abundance, and Proposition 53 produces a 3-parameter family of size-k pseudocontexts whenever the chosen frame has finite ray-stabiliser. Among the most symmetric examples we have the Platonic frames and their pyramid generalisations:

- $k = 4$: regular tetrahedron, $T = \frac{4}{3} I_3$ (Example 58);
- $k = 5$: a regular square pyramid whose four base vectors are elevated to angle $\arccos(1/\sqrt{6})$ from the apex, $T = \frac{5}{3} I_3$;
- $k = 6$: one representative from each of the six antipodal vertex pairs of a regular icosahedron, $T = 2I_3$;
- $k = 7$: a regular hexagonal pyramid whose six base vectors are elevated to angle $\arccos(\sqrt{2}/3)$ from the apex, $T = \frac{7}{3} I_3$;
- $k = 12$: a union of three generically rotated tetrahedral tight frames, chosen with no ray coincidences, gives $T = 4I_3$.

While Platonic solids naturally furnish perfectly symmetric (equiangular) frames for $k = 4$ and $k = 6$, the cases of $k = 5$ and $k = 7$ illustrate a fundamental structural phase shift. By Gerzon's bound, the maximum number of vectors in a real equiangular tight frame (ETF) in $\mathbb{R}^n$ is $\frac{1}{2} n(n+1)$, which caps at exactly $k = 6$ for $\mathbb{R}^3$. Furthermore, real ETFs only exist for restricted sizes (namely $k = 3, 4, 6$ in $\mathbb{R}^3$). Consequently, no strictly maximally symmetric configurations can exist for $k = 5$ or $k \geq 7$.

However, to compensate for the loss of perfect symmetry, these non-ETF frames acquire internal degrees of freedom. The expected local parameter count for the unit-norm tight-frame variety in $\mathbb{R}^3$, modulo rotations and away from singular points, is $2k - 8$. Thus, while $k = 4$ represents a rigid shape, the highly symmetric "square pyramid" of $k = 5$ sits inside a positive-dimensional space of tight-frame deformations (e.g., stretching into rectangular or rhombic pyramids), and the $k = 7$ configuration has still more deformation directions. Each point in such a configuration space generates its own 3-parameter rotational family of real pseudocontexts spanning $\mathbb{R}^3$.

Remark 67. Together with the planar Mercedes–Benz construction (Example 20) and the nondegenerate $k = 3$ realisations in $\mathbb{R}^3$ of [4] (cf. Remark 23), Example 58 completes a coherent picture of real pseudocontexts in low ambient dimension: every $k \geq 3$ admits genuine real realisations spanning $\mathbb{R}^3$. Those for $k = 3$ are necessarily nondegenerate (Corollary 66), those for $k = 4$ and $k = 6$ admit highly rigid, rotationally parametrised ETF families, and those for $k \geq 5$ (lacking ETF structure) exist within continuous, multidimensional spaces of internally deforming tight realisations. Numerical continuation confirms that unconstrained, purely vectorial $k = 3$ pseudocontexts also exist in continuous deformations (e.g., with generic spectra such as $\lambda \approx \{0.68, 1.16, 1.16\} \approx (1/7)\{5, 8, 8\}$), though they lack the global rotational symmetry of the $k \geq 4$ tight frames.

Example 68 (Pyramidal tight frames for $k = 5$ and $k = 7$)**.** By Gerzon's bound [13, Theorem 3.5], no real equiangular tight frame (ETF) can contain more than $\frac{1}{2} n(n+1)$ vectors, meaning no perfectly symmetric Platonic configuration can exist in $\mathbb{R}^3$ for $k = 5$ or $k \geq 7$. Nonetheless, highly symmetric unit-norm tight frames can be constructed via regular pyramids. For $k = 5$, a tight frame with $T = \frac{5}{3} I_3$ is formed by a square pyramid with apex $v_1 = (0, 0, 1)$ and four base vectors at an angle of $\arccos(1/\sqrt{6})$ from the apex:

$$v_{2,3} = \left(\pm \sqrt{\frac{5}{6}}, 0, \frac{1}{\sqrt{6}} \right), \quad v_{4,5} = \left(0, \pm \sqrt{\frac{5}{6}}, \frac{1}{\sqrt{6}} \right).$$

Similarly, for $k = 7$, a tight frame with $T = \frac{7}{3} I_3$ is given by a hexagonal pyramid with apex $v_1 = (0, 0, 1)$ and six base vectors at an angle of $\arccos(\sqrt{2}/3)$ from the apex:

$$v_{j+2} = \left(\frac{\sqrt{7}}{3} \cos \frac{j\pi}{3}, \frac{\sqrt{7}}{3} \sin \frac{j\pi}{3}, \frac{\sqrt{2}}{3} \right) \quad \text{for } j \in \{0, \dots, 5\}.$$

Unlike the rigid ETFs at $k = 4$ and $k = 6$, the expected local parameter count for the unit-norm tight-frame equations in $\mathbb{R}^3$ is $2k - 8$ after quotienting by rotations. Thus these $k = 5$ and $k = 7$ pyramids should be viewed as highly symmetric points inside positive-dimensional spaces of tight realisations, rather than as isolated Platonic configurations.

XII. OPEN PROBLEMS

Open Problem 69 (Classification over $\mathbb{C}^n$). Classify, up to unitary equivalence, all pseudocontexts of size k in $\mathbb{C}^n$. Even for small (k,n), the orbit structure is not yet transparent.

Open Problem 70 (Lattice representability). Characterise which orthomodular lattices carrying a pseudocontext-type additive identification admit a faithful embedding into some $L(\mathbb{C}^n)$, sharpening the partial results of [4, 14].

Open Problem 71 (Frame-symmetry characterisation in the partially degenerate case). For a positive operator T with spectral decomposition

$$T = \sum_\alpha \lambda_\alpha P_\alpha,$$

describe the decomposition space

$$\mathscr{D}(T) = \left\{ (a_1,\ldots,a_k) : \sum_i P_{a_i} = T,\ \|a_i\| = 1 \right\}$$

modulo column phases, permutations, and the commutant

$$\prod_\alpha U(\operatorname{rank} P_\alpha).$$

In particular, determine the commutant-orbit strata and the residual moduli of inequivalent decompositions for partially degenerate spectra.

Open Problem 72 (Hypergraph-forced higher-dimensional pseudocontexts). The rotation construction of Section XI produces continuous families of operator-level pseudocontexts spanning all of R^n. In generic examples, such as the tetrahedral family, there is strong evidence that the identity is not graph-forced by a finite orthogonality hypergraph. In contrast, in [4, 14] and in Section IX of the present paper, the orthogonality relations themselves induce condition (iii) of Definition 3 for every vector representation fulfilling the orthogonality constraints. The systematic classification of hypergraphs that force a pseudocontext identity, and of the higher-dimensional configurations they admit, remains open. This stratum is disjoint from the one resolved in Section XI: the tetrahedral example is not a graph-forced pseudocontext, since the topological covering technique of Section VIII cannot be applied to it.

Open Problem 73 (Classification of minimal real pseudocontexts). Theorem 12(ii) and the proposition that follows it (Proposition 14) establish that genuine real pseudocontexts of size $k=2$ do not exist in any dimension, while Example 20 demonstrates their existence for $k=3$ in two-dimensional subspaces and [4] exhibits $\mathbb{R}^3$-spanning instances. Thus $k=3$ is the minimal size for genuine real pseudocontexts. Classify all such minimal ($k=3$) examples up to orthogonal equivalence over $\mathbb{R}$, distinguishing the two-dimensional case (where the doubled-angle projective-circle description reduces the problem to equal-centroid decompositions of three unit vectors, with scalar structure operators only in the zero-centroid tight-frame subcase) from the genuinely three-dimensional case (where, by Corollary 66, the structure operator is necessarily nondegenerate).

Open Problem 74 (Moduli of tight real pseudocontexts). For each $k \geq n+1$, describe the moduli space $\mathscr{M}_{k,n}^{\text{tight}} = \left\{\text{unit-norm tight frames of size } k \text{ in } \mathbb{R}^n\right\} / O(n) \times S_k$, where S_k acts by permutation. Together with Proposition 53, an explicit description of $\mathscr{M}_{k,n}^{\text{tight}}$ would yield a complete parametrisation of the tight stratum of real $\mathbb{R}^n$-spanning pseudocontexts of size k. The case $(k,n) = (4,3)$ reduces to a point (the regular tetrahedron, by Example 58); $(k,n) = (5,3)$ already gives a positive-dimensional moduli space.

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Appendix A: Derivation of the Master Constraint

We provide a step-by-step verification of the algebraic derivation leading from the perimeter closure conditions to the master constraint (13) used in the proof of Proposition 49.

The starting point is the perimeter closure condition stated in the proof: for three rays v_{j-1}, v_j, v_{j+1} forming a context, adjacent corner rays are orthogonal, yielding

$$(v_{j-1}\cdot v_j)(v_j\cdot v_{j+1}) - (v_{j-1}\cdot v_{j+1})\|v_j\|^2 = 0. \tag{A1}$$

Evaluating (A1) at $j=0$ and $j=4$ produces the relations

$$Y_0 = \frac{Y_7 + mX_1}{1 - mX_1Y_7}, \tag{A2}$$

$$Y_4 = \frac{Y_3 + mX_5}{1 - mY_3X_5}, \tag{A3}$$

as stated in the proof. The inner contexts impose the constraints

$$Y_7 = -\frac{1}{MY_3}, \qquad X_5 = -\frac{1}{MX_1}, \qquad M = 1+m^2, \tag{A4}$$

together with the requirement $Y_0Y_4 = -1$ from context D_0.

Simplifying the fractions

Substituting Y_7 from (A4) into (A2):

$$Y_0 = \frac{-\frac{1}{MY_3} + mX_1}{1 - mX_1\left(-\frac{1}{MY_3}\right)} = \frac{mMX_1Y_3 - 1}{MY_3 + mX_1}. \tag{A5}$$

Substituting X_5 from (A4) into (A3):

$$Y_4 = \frac{Y_3 + m\left(-\frac{1}{MX_1}\right)}{1 - mY_3\left(-\frac{1}{MX_1}\right)} = \frac{MX_1Y_3 - m}{MX_1 + mY_3}. \tag{A6}$$

Constructing the product condition

Let $u = X_1Y_3$. Using the constraint $Y_0Y_4 = -1$ from context D_0 together with (A5) and (A6):

$$\frac{mMu-1}{mX_1+MY_3}\cdot\frac{Mu-m}{MX_1+mY_3} = -1. \tag{A7}$$

Cross-multiplying (A7) yields

$$(mMu-1)(Mu-m) = -(mX_1+MY_3)(MX_1+mY_3). \tag{A8}$$

Expanding both sides

The left-hand side of (A8) expands as

$$\begin{aligned}\text{LHS} &= mM^2u^2 - m^2Mu - Mu + m\\ &= mM^2u^2 - Mu(m^2+1) + m.\end{aligned} \tag{A9}$$

Since $M = 1+m^2$, the coefficient m^2+1 equals M, giving

$$\text{LHS} = mM^2u^2 - M^2u + m. \tag{A10}$$

The right-hand side of (A8) expands as

$$\begin{aligned}\text{RHS} &= -\left[mMX_1^2 + m^2X_1Y_3 + M^2X_1Y_3 + mMY_3^2\right]\\ &= -\left[u(M^2+m^2) + mM(X_1^2+Y_3^2)\right].\end{aligned} \tag{A11}$$

Cancellation and the master constraint

Equating (A10) and (A11):

$$mM^2u^2 - M^2u + m = -u(M^2+m^2) - mM(X_1^2+Y_3^2). \tag{A12}$$

Adding M^2u to both sides, the $-M^2u$ term cancels:

$$mM^2u^2 + m = -m^2u - mM(X_1^2+Y_3^2). \tag{A13}$$

Bringing all terms to the left and dividing by $m \neq 0$ (since $\theta \notin \frac{\pi}{2}\mathbb{Z}$ by faithfulness) yields the master constraint (13) in the proof of Proposition 49.

Positivity contradiction

Since $X_1, Y_3 \in \mathbb{R}$, the inequality $X_1^2 + Y_3^2 \geq 2|X_1Y_3| = 2|u|$ holds. The left-hand side of (13) therefore satisfies

$$\begin{aligned}&M^2u^2 + mu + 1 + M(X_1^2+Y_3^2)\\ &\geq M^2u^2 + mu + 1 + 2M|u|\\ &\geq M^2u^2 + |u|(2M-|m|) + 1.\end{aligned} \tag{A14}$$

With $M = 1+m^2 > 1$, we have $2M - |m| = 2 + 2m^2 - |m| > 0$ for all real m. Consequently the expression in (A14) is strictly positive for every $u \in \mathbb{R}$, contradicting (13). This completes the algebraic verification of the master constraint and the resulting impossibility of a real realization.

Appendix B: Set representation of the octagon hypergraph via two-valued states

As described in Section 2.3 of Ref. [15], any finite atomic logic equipped with a separating set of dispersion-free probability measures (two-valued states) can be faithfully represented as a partition logic [7–9]. In this formulation, the set representation is constructed by "inverting" the set of two-valued states: every vertex v (atom) of the hypergraph is mapped to the set of indices of those two-valued states that evaluate to 1 ("true") on it. Explicitly, if M denotes the complete set of two-valued states s_k, the set representation of a vertex v is given by

$$p(v) = \{k \mid s_k(v) = 1, k \in M\}. \tag{B1}$$

Under this mapping, every context (orthogonal basis) translates into a disjoint partition of the total state space M.

1. Representation of the incomplete configuration

An exhaustive enumeration of the two-valued states on the tight pseudocontext configuration introduced in Section IX (the incomplete octagon hypergraph depicted in Fig. 2) reveals exactly 26 such states. Because this set of 26 states mutually separates every pair of vertices, the resulting set representation is faithful.

Let us index the set of two-valued states as $M_{26} = \{1, 2, \ldots, 26\}$. The full evaluation grid—referred to here as the Travis matrix [8, 16]—is presented in Table II, where the states s_k act as the rows and the vertices as the columns. By taking the vertical union of the evaluated indices where $s_k(v) = 1$, we obtain the subsets representing each vertex. Traversing the 16 perimeter vertices in counterclockwise order starting from v_{56}, followed by the 4 inner vertices a_1, b_1, a_2, b_2, the explicit mapping is as follows:

$$\begin{aligned}
p(v_{56}) &= \{1,2,3,4,5,6,7,8,9\},\\
p(v_6) &= \{10,11,12,13,14,15,16,17\},\\
p(v_{67}) &= \{18,19,20,21,22,23,24,25,26\},\\
p(v_7) &= \{1,2,3,4,10,11,12,13\},\\
p(v_{70}) &= \{5,6,7,8,9,14,15,16,17\},\\
p(v_0) &= \{1,2,10,11,18,19,20,21\},\\
p(v_{01}) &= \{3,4,12,13,22,23,24,25,26\},\\
p(v_1) &= \{1,5,6,7,10,14,18,19\},\\
p(v_{12}) &= \{2,8,9,11,15,16,17,20,21\},\\
p(v_2) &= \{1,3,5,6,18,22,23,24\},\\
p(v_{23}) &= \{4,7,10,12,13,14,19,25,26\},\\
p(v_3) &= \{5,8,15,16,18,20,22,23\},\\
p(v_{34}) &= \{1,2,3,6,9,11,17,21,24\},\\
p(v_4) &= \{4,5,7,8,12,15,22,25\},\\
p(v_{45}) &= \{10,13,14,16,18,19,20,23,26\},\\
p(v_5) &= \{11,12,15,17,21,22,24,25\},
\end{aligned}$$

$$\begin{aligned}
p(a_1) &= \{3,6,9,13,14,16,17,23,24,26\},\\
p(b_1) &= \{6,7,9,14,17,19,21,24,25,26\},\\
p(a_2) &= \{2,4,7,8,9,19,20,21,25,26\},\\
p(b_2) &= \{2,3,4,8,9,13,16,20,23,26\}.
\end{aligned}$$

2. Representation of the full configuration

When considering the full context configuration of Fig. 3, which includes the additional intertwining point c and two more contexts ($\{a_1,c,a_2\}$ and $\{b_1,c,b_2\}$), the tighter hypergraph imposes stronger constraints and yields a set of exactly 24 two-valued states.

Let us index this new state space as $M_{24} = \{1,2,\ldots,24\}$. The Travis matrix has dimensions 24×21, corresponding to the 24 states and the 20 previous vertices plus the new vertex c (see Table III). Extracting the non-zero indices column by column provides the explicit set representation for the full hypergraph:

$$\begin{aligned}
p(v_{56}) &= \{1,2,3,4,5,6,7,8\},\\
p(v_6) &= \{9,10,11,12,13,14,15,16\},\\
p(v_{67}) &= \{17,18,19,20,21,22,23,24\},\\
p(v_7) &= \{1,2,3,4,9,10,11,12\},\\
p(v_{70}) &= \{5,6,7,8,13,14,15,16\},\\
p(v_0) &= \{1,2,9,10,17,18,19,20\},\\
p(v_{01}) &= \{3,4,11,12,21,22,23,24\},\\
p(v_1) &= \{1,5,6,7,9,13,17,18\},\\
p(v_{12}) &= \{2,8,10,14,15,16,19,20\},\\
p(v_2) &= \{1,3,5,6,17,21,22,23\},\\
p(v_{23}) &= \{4,7,9,11,12,13,18,24\},\\
p(v_3) &= \{5,8,14,15,17,19,21,22\},\\
p(v_{34}) &= \{1,2,3,6,10,16,20,23\},\\
p(v_4) &= \{4,5,7,8,11,14,21,24\},\\
p(v_{45}) &= \{9,12,13,15,17,18,19,22\},\\
p(v_5) &= \{10,11,14,16,20,21,23,24\},
\end{aligned}$$

$$\begin{aligned}
p(a_1) &= \{3,6,12,13,15,16,22,23\},\\
p(b_1) &= \{6,7,13,16,18,20,23,24\},\\
p(a_2) &= \{2,4,7,8,18,19,20,24\},\\
p(b_2) &= \{2,3,4,8,12,15,19,22\},
\end{aligned}$$

$$p(c) = \{1,5,9,10,11,14,17,21\}.$$

It is straightforward to verify that every context, including the newly added ones, maps to a valid disjoint partition of the complete state space M_{24}. For example, the new context $\{a_1,c,a_2\}$ yields the exact partition $p(a_1)\cup p(c)\cup p(a_2) = M_{24}$. This explicit equivalence confirms that the logical skeleton of the tight pseudocontext is fully robust, supporting a classical, mutually separating set representation of automaton logic alongside its quantum representations.

3. Comparison of deterministic, classical, and quantum probability sums

The indicator or characteristic function $\mathbf{1}_A(x) = 1$ if $x\in A$ and 0 if $x\notin A$. For a vertex v, write

$$s_i(v) = \mathbf{1}_{p(v)}(i),$$

where $p(v) = \{i : s_i(v) = 1\}$ is its support in the Travis representation.

The equality

$$p(a_1)\cup p(a_2) = p(b_1)\cup p(b_2)$$

therefore says only that the two pairs have the same support for the corresponding logical OR. It does not, by itself, imply equality of the arithmetic sums of truth values. Indeed, for two Boolean variables x,y,

$$x+y = (x\vee y)+(x\wedge y).$$

TABLE II. The Travis matrix of the 26 two-valued states for the incomplete octagon configuration. Rows $s_1, \ldots, s_{26}$ denote the states, and columns represent the vertices.

State	v_{56}	v_6	v_{67}	v_7	v_{70}	v_0	v_{01}	v_1	v_{12}	v_2	v_{23}	v_3	v_{34}	v_4	v_{45}	v_5	a_1	b_1	a_2	b_2
s_1	1	0	0	1	0	1	0	1	0	1	0	0	1	0	0	0	0	0	0	0
s_2	1	0	0	1	0	1	0	0	1	0	0	0	1	0	0	0	0	0	1	1
s_3	1	0	0	1	0	0	1	0	0	1	0	0	1	0	0	0	1	0	0	1
s_4	1	0	0	1	0	0	1	0	0	0	1	0	0	1	0	0	0	0	1	1
s_5	1	0	0	0	1	0	0	1	0	1	0	1	0	1	0	0	0	0	0	0
s_6	1	0	0	0	1	0	0	1	0	1	0	0	1	0	0	0	1	1	0	0
s_7	1	0	0	0	1	0	0	1	0	0	1	0	0	1	0	0	0	1	1	0
s_8	1	0	0	0	1	0	0	0	1	0	0	1	0	1	0	0	0	0	1	1
s_9	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	1	1	1
s_{10}	0	1	0	1	0	1	0	1	0	0	1	0	0	0	1	0	0	0	0	0
s_{11}	0	1	0	1	0	1	0	0	1	0	0	0	1	0	0	1	0	0	0	0
s_{12}	0	1	0	1	0	0	1	0	0	0	1	0	0	1	0	1	0	0	0	0
s_{13}	0	1	0	1	0	0	1	0	0	0	1	0	0	0	1	0	1	0	0	1
s_{14}	0	1	0	0	1	0	0	1	0	0	1	0	0	0	1	0	1	1	0	0
s_{15}	0	1	0	0	1	0	0	0	1	0	0	1	0	1	0	1	0	0	0	0
s_{16}	0	1	0	0	1	0	0	0	1	0	0	1	0	0	1	0	1	0	0	1
s_{17}	0	1	0	0	1	0	0	0	1	0	0	0	1	0	0	1	1	1	0	0
s_{18}	0	0	1	0	0	1	0	1	0	1	0	1	0	0	1	0	0	0	0	0
s_{19}	0	0	1	0	0	1	0	1	0	0	1	0	0	0	1	0	0	1	1	0
s_{20}	0	0	1	0	0	1	0	0	1	0	0	1	0	0	1	0	0	0	1	1
s_{21}	0	0	1	0	0	1	0	0	1	0	0	0	1	0	0	1	0	1	1	0
s_{22}	0	0	1	0	0	0	1	0	0	1	0	1	0	1	0	1	0	0	0	0
s_{23}	0	0	1	0	0	0	1	0	0	1	0	1	0	0	1	0	1	0	0	1
s_{24}	0	0	1	0	0	0	1	0	0	1	0	0	1	0	0	1	1	1	0	0
s_{25}	0	0	1	0	0	0	1	0	0	0	1	0	0	1	0	1	0	1	1	0
s_{26}	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	1	1	1	1

TABLE III. The Travis matrix of the 24 two-valued states for the full octagon configuration. Rows $s_1, \ldots, s_{24}$ denote the states, and columns represent the vertices.

State	v_{56}	v_6	v_{67}	v_7	v_{70}	v_0	v_{01}	v_1	v_{12}	v_2	v_{23}	v_3	v_{34}	v_4	v_{45}	v_5	a_1	b_1	a_2	b_2	c
s_1	1	0	0	1	0	1	0	1	0	1	0	0	1	0	0	0	0	0	0	0	1
s_2	1	0	0	1	0	1	0	0	1	0	0	0	1	0	0	0	0	0	1	1	0
s_3	1	0	0	1	0	0	1	0	0	1	0	0	1	0	0	0	1	0	0	1	0
s_4	1	0	0	1	0	0	1	0	0	0	1	0	0	1	0	0	0	0	1	1	0
s_5	1	0	0	0	1	0	0	1	0	1	0	1	0	1	0	0	0	0	0	0	1
s_6	1	0	0	0	1	0	0	1	0	1	0	0	1	0	0	0	1	1	0	0	0
s_7	1	0	0	0	1	0	0	1	0	0	1	0	0	1	0	0	0	1	1	0	0
s_8	1	0	0	0	1	0	0	0	1	0	0	1	0	1	0	0	0	0	1	1	0
s_9	0	1	0	1	0	1	0	1	0	0	1	0	0	0	1	0	0	0	0	0	1
s_{10}	0	1	0	1	0	1	0	0	1	0	0	0	1	0	0	1	0	0	0	0	1
s_{11}	0	1	0	1	0	0	1	0	0	0	1	0	0	1	0	1	0	0	0	0	1
s_{12}	0	1	0	1	0	0	1	0	0	0	1	0	0	0	1	0	1	0	0	1	0
s_{13}	0	1	0	0	1	0	0	1	0	0	1	0	0	0	1	0	1	1	0	0	0
s_{14}	0	1	0	0	1	0	0	0	1	0	0	1	0	1	0	1	0	0	0	0	1
s_{15}	0	1	0	0	1	0	0	0	1	0	0	1	0	0	1	0	1	0	0	1	0
s_{16}	0	1	0	0	1	0	0	0	1	0	0	0	1	0	0	1	1	1	0	0	0
s_{17}	0	0	1	0	0	1	0	1	0	1	0	1	0	0	1	0	0	0	0	0	1
s_{18}	0	0	1	0	0	1	0	1	0	0	1	0	0	0	1	0	0	1	1	0	0
s_{19}	0	0	1	0	0	1	0	0	1	0	0	1	0	0	1	0	0	0	1	1	0
s_{20}	0	0	1	0	0	1	0	0	1	0	0	0	1	0	0	1	0	1	1	0	0
s_{21}	0	0	1	0	0	0	1	0	0	1	0	1	0	1	0	1	0	0	0	0	1
s_{22}	0	0	1	0	0	0	1	0	0	1	0	1	0	0	1	0	1	0	0	1	0
s_{23}	0	0	1	0	0	0	1	0	0	1	0	0	1	0	0	1	1	1	0	0	0
s_{24}	0	0	1	0	0	0	1	0	0	0	1	0	0	1	0	1	0	1	1	0	0

Equivalently, for indicator (characteristic) functions,

$$\mathbf{1}_A + \mathbf{1}_B = \mathbf{1}_{A\cup B} + \mathbf{1}_{A\cap B}.$$

Thus equality of unions must be supplemented by equality of intersections.

In the 26-state representation one finds

$$p(a_1) \cap p(a_2) = p(b_1) \cap p(b_2) = \{9, 26\},$$

whereas in the 24-state representation one finds

$$p(a_1) \cap p(a_2) = p(b_1) \cap p(b_2) = \varnothing.$$

Together with the corresponding union equalities, this implies that, in these two particular configurations,

$$s_i(a_1)+s_i(a_2)=s_i(b_1)+s_i(b_2)$$

for every two-valued state s_i in the complete enumeration. This pointwise identity should be understood as a property verified from the full Travis matrix, or derived from the context-covering equations of Section IX, not as a consequence of union equality alone.

For deterministic two-valued states, the four-term sum

$$s_i(a_1)+s_i(a_2)+s_i(b_1)+s_i(b_2)=2\left[s_i(a_1)+s_i(a_2)\right]$$

therefore takes the values $0,2,4$ in the incomplete case, but only $0,2$ in the full $C=1$ configuration. For classical hidden-variable mixtures over the enumerated two-valued states,

$$P_{\rm cl}(v)=\sum_i \lambda_i s_i(v), \qquad \lambda_i\geq 0, \qquad \sum_i \lambda_i=1,$$

the four-term sum is no longer restricted to these discrete values, but ranges over their convex hull: $[0,4]$ in the incomplete case and $[0,2]$ in the full $C=1$ case.

On the quantum side, the corresponding equality is expressed by the structure operator

$$T=P_{a_1}+P_{a_2}=P_{b_1}+P_{b_2}.$$

Hence

$$P_Q(a_1)+P_Q(a_2)+P_Q(b_1)+P_Q(b_2)=2\,\mathrm{tr}(\rho T).$$

For a two-dimensional support V, the $k=2$ Bloch representation gives

$$T=I_V+\frac{1}{2}\vec{R}\cdot\vec{\sigma},$$

with eigenvalues

$$\lambda_\pm=1\pm\frac{1}{2}\|\vec{R}\|.$$

Thus the maximal quantum four-term sum is

$$2\lambda_{\max}(T)=2+\|\vec{R}\|.$$

In the incomplete nonorthogonal case, $\vec{R}\neq 0$, so this maximum is larger than 2, but still strictly below the deterministic classical maximum 4 unless the two rays collapse. In the full $C=1$ case, orthogonality forces $\vec{R}=0$, so

$$T=I_V.$$

Since $\{a_1,c,a_2\}$ is a context in the full $C=1$ configuration, $P_{a_1}+P_c+P_{a_2}=I$, giving $T=I-P_c$. Consequently,

$$0\leq 2\,\mathrm{tr}(\rho T)\leq 2,$$

with equality 2 for states supported entirely on $V=c^\perp$. Thus the added contexts remove the deterministic classical excess 4 and bring the classical convex bound for the four-term sum into agreement with the quantum spectral bound.

4. Faithfulness and the Kochen–Specker distinction

The faithfulness of the two set representations above is established directly by the complete Travis matrices: distinct vertices have distinct support sets $p(v)$, and every context is represented as a disjoint partition of the relevant state space M_{26} or M_{24}. Hence these particular hypergraphs admit faithful partition-logic representations.

This property should not be understood as automatic for arbitrary finite orthogonality hypergraphs. Kochen–Specker configurations provide the opposite phenomenon: the orthogonality constraints may be so strong that no separating family of globally consistent two-valued states exists, and in some cases no two-valued state exists at all. Thus the present octagon configurations lie on the classically representable side of the Zierler–Schlessinger/Greechie criterion, whereas Kochen–Specker sets lie on the nonrepresentable side.

[1] R. Mayet, M. Navara, and V. Rogalewicz, Orthomodular lattices with rich state spaces, Algebra Universalis **43**, 1 (2000).

[2] A. M. Gleason, Measures on the closed subspaces of a Hilbert space, Journal of Mathematics and Mechanics (now Indiana University Mathematics Journal) **6**, 885 (1957).

[3] M. Navara and K. Svozil, Pseudocontexts, Presentation at *Quantum Structures 2024* (2024), July 1–5, 2024.

[4] M. Navara and K. Svozil, Form of contextuality predicting probabilistic equivalence between two sets of three mutually noncommuting observables, Physical Review A **109**, 022222 (2024), arXiv:2309.13091.

[5] L. Lovász, On the Shannon capacity of a graph, IEEE Transactions on Information Theory **25**, 1 (1979).

[6] A. Bretto, *Hypergraph Theory*, Mathematical Engineering (Springer, Cham, Switzerland, 2013) pp. xiv+119.

[7] N. Zierler and M. Schlessinger, Boolean embeddings of orthomodular sets and quantum logic, Duke Mathematical Journal **32**, 251 (1965), reprinted in Ref. [17].

[8] R. J. Greechie, *Orthomodular Lattices*, Ph.D. thesis, University of Florida, Florida, USA (1966).

[9] S. Kochen and E. P. Specker, The problem of hidden variables in quantum mechanics, Journal of Mathematics and Mechanics (now Indiana University Mathematics Journal) **17**, 59 (1967).

[10] J. Harding and R. Salinas Schmeis, Remarks on orthogonality spaces, International Journal of Theoretical Physics **64**, 211 (2025), arXiv:2505.13871 [math-ph], arXiv:arXiv:2505.13871 [math-ph].

[11] A. Cabello, J. M. Estebaranz, and G. García-Alcaine, Bell-Kochen-Specker theorem: A proof with 18 vectors, Physics Letters A **212**, 183 (1996), arXiv:quant-ph/9706009.

[12] R. W. Spekkens, Contextuality for preparations, transformations, and unsharp measurements, Physical Review A **71**,

052108 (2005), arXiv:quant-ph/0406166.

[13] P. W. H. Lemmens and J. J. Seidel, Equiangular lines, Journal of Algebra **24**, 494 (1973).

[14] M. Navara and K. Svozil, Exploring quantum contextuality with the quantum Möbius-Escher-Penrose hypergraph, Physical Review A **111**, 042209 (2025).

[15] K. Svozil, Logical equivalence between generalized urn models and finite automata, International Journal of Theoretical Physics **44**, 745 (2005), arXiv:quant-ph/0209136.

[16] R. D. Travis, *The Logic of a Physical Theory*, Master's thesis, Wayne State University, Detroit, Michigan, USA (1962), Master's Thesis under the supervision of David J. Foulis.

[17] N. Zierler and M. Schlessinger, Boolean embeddings of orthomodular sets and quantum logic, in *The Logico-Algebraic Approach to Quantum Mechanics: Volume I: Historical Evolution*, edited by C. A. Hooker (Springer Netherlands, Dordrecht, The Netherlands, 1975) pp. 247–262.